

# Memory Scaling: 大模型持续学习机制探索

汇报人

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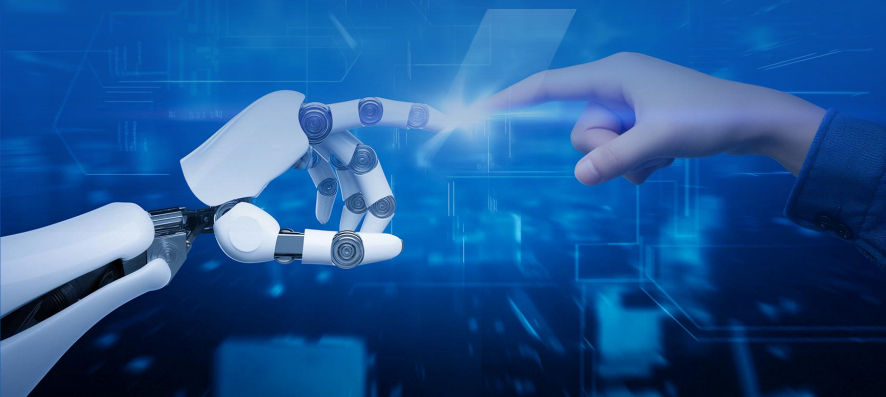
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# 主要内容

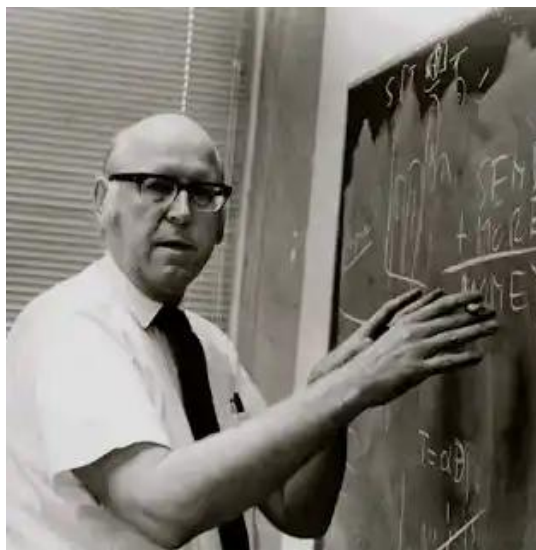
**01 | 大模型记忆智能**

02 | 智能体记忆工作

03 | 相关工作与展望

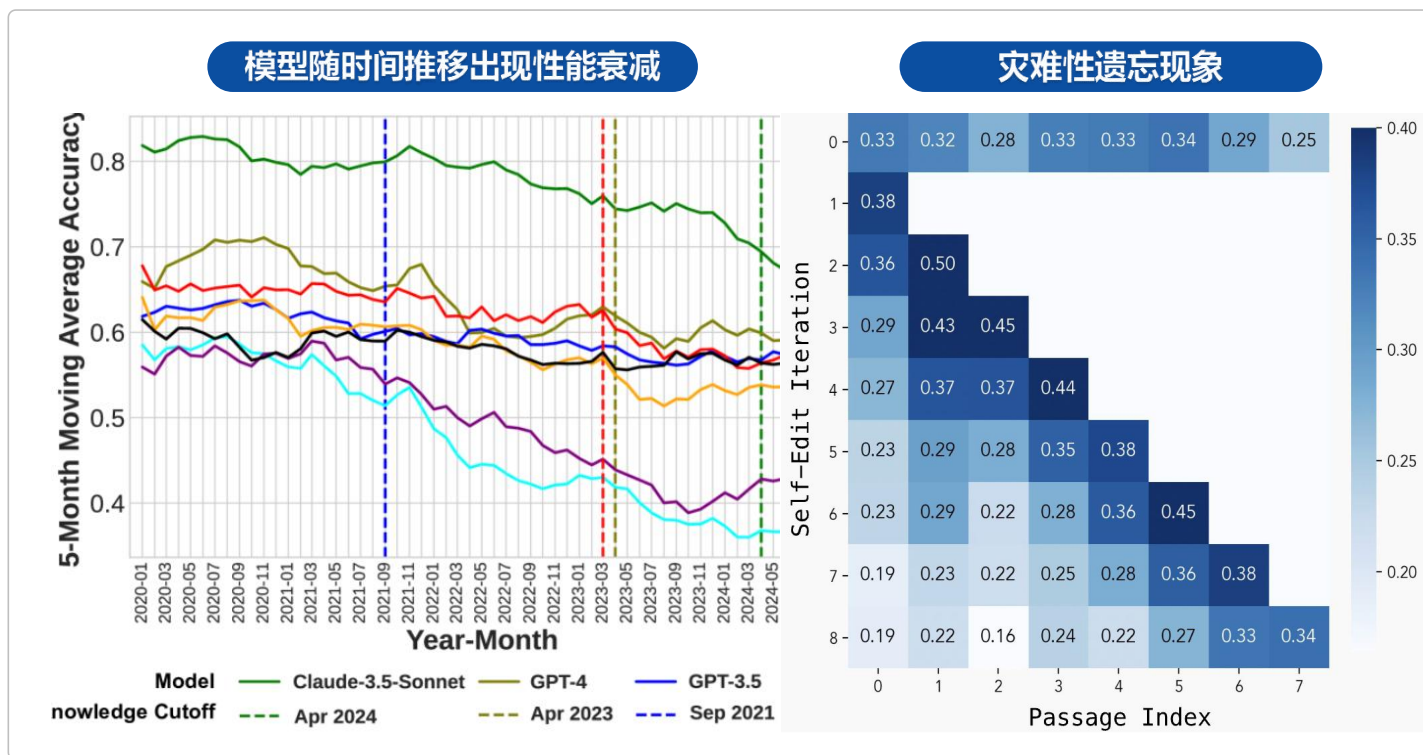


- 开放世界环境是**动态**和**演进**的，正不断涌现新概念、新现象
- 大模型必须克服其**静态**与**固化**局限，实现与环境的同步演化



“

人工智能先驱、图灵奖获得者Allen Newell 教授、  
司马贺教授将智能定义为**实现自身目的并适应环境**



- 人类智能中，记忆是**动态的生成与重构的过程**，对环境信息进行深入**抽象、关联与整合**。记忆是实现持续、连贯和个性化交互等高级认知的核心基石

## 记忆的三个环节

## 编码

外部信息重组、转化

## 组织

分层分类保存长期记忆

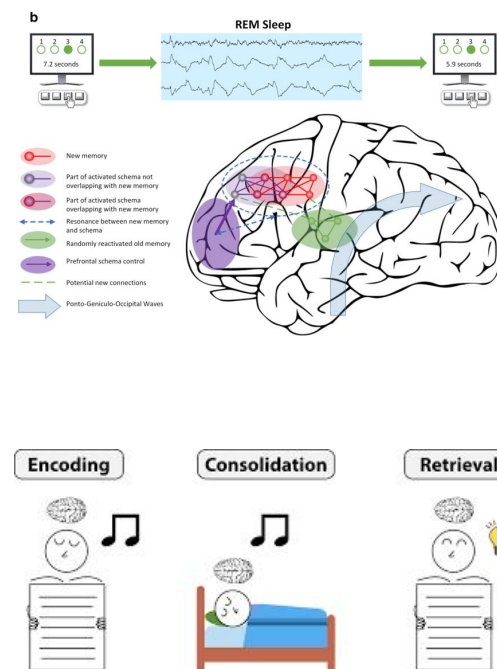
## 检索

从长期记忆中提取信息

➤ **编码：**学习效率取决于记忆编码策略。多通道编码、情景联想等策略显著提升学习效果

➤ **组织：**保留所学内容，使得后续学习更高效

➤ **检索：**巩固记忆存储，激发元认知能力，促进推理、抽象与联想

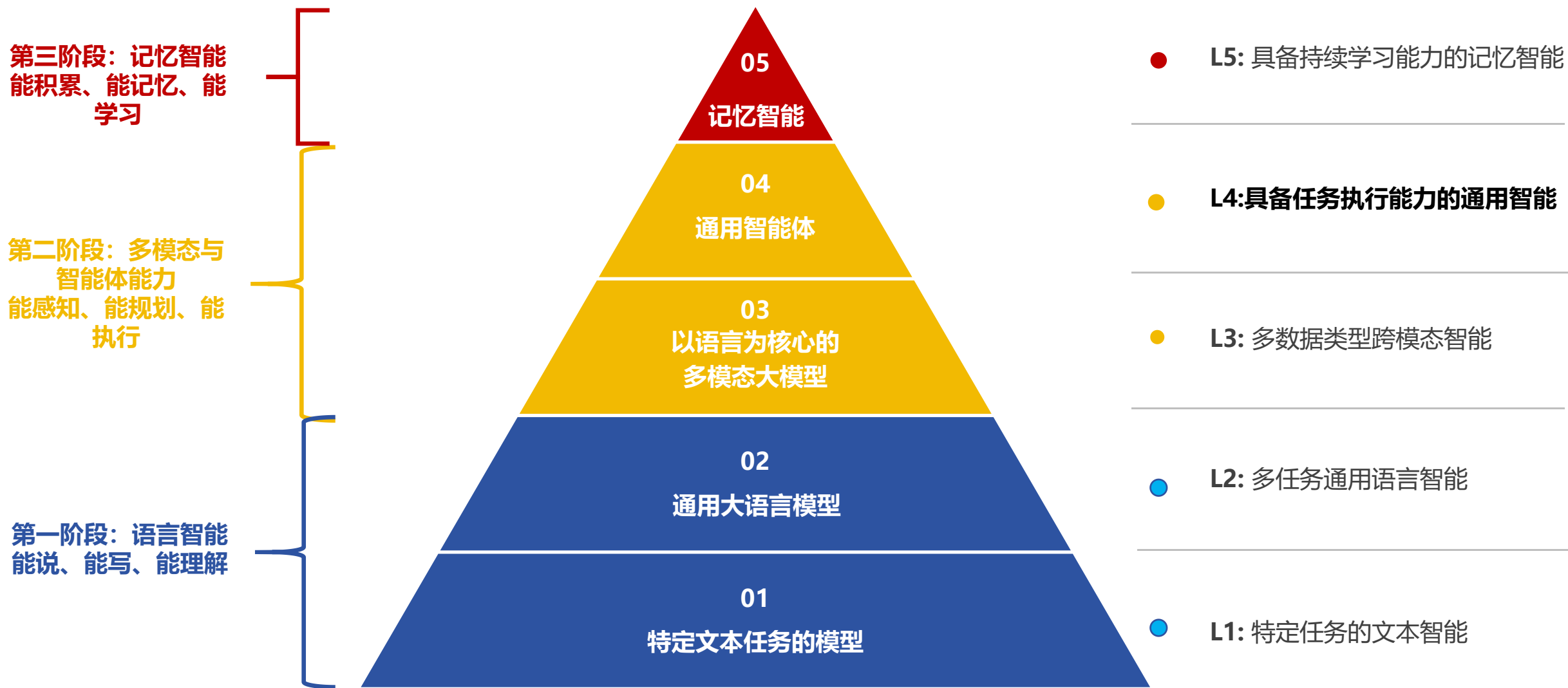


Pyc, M. A., & Rawson, K. A. (2010). Why testing improves memory: Mediator effectiveness hypothesis. *Science*, 330(6002), 335-335.

Bjork R A. (1994). Memory and Metamemory Considerations in the. *Metacognition: Knowing about Knowing*, 185.

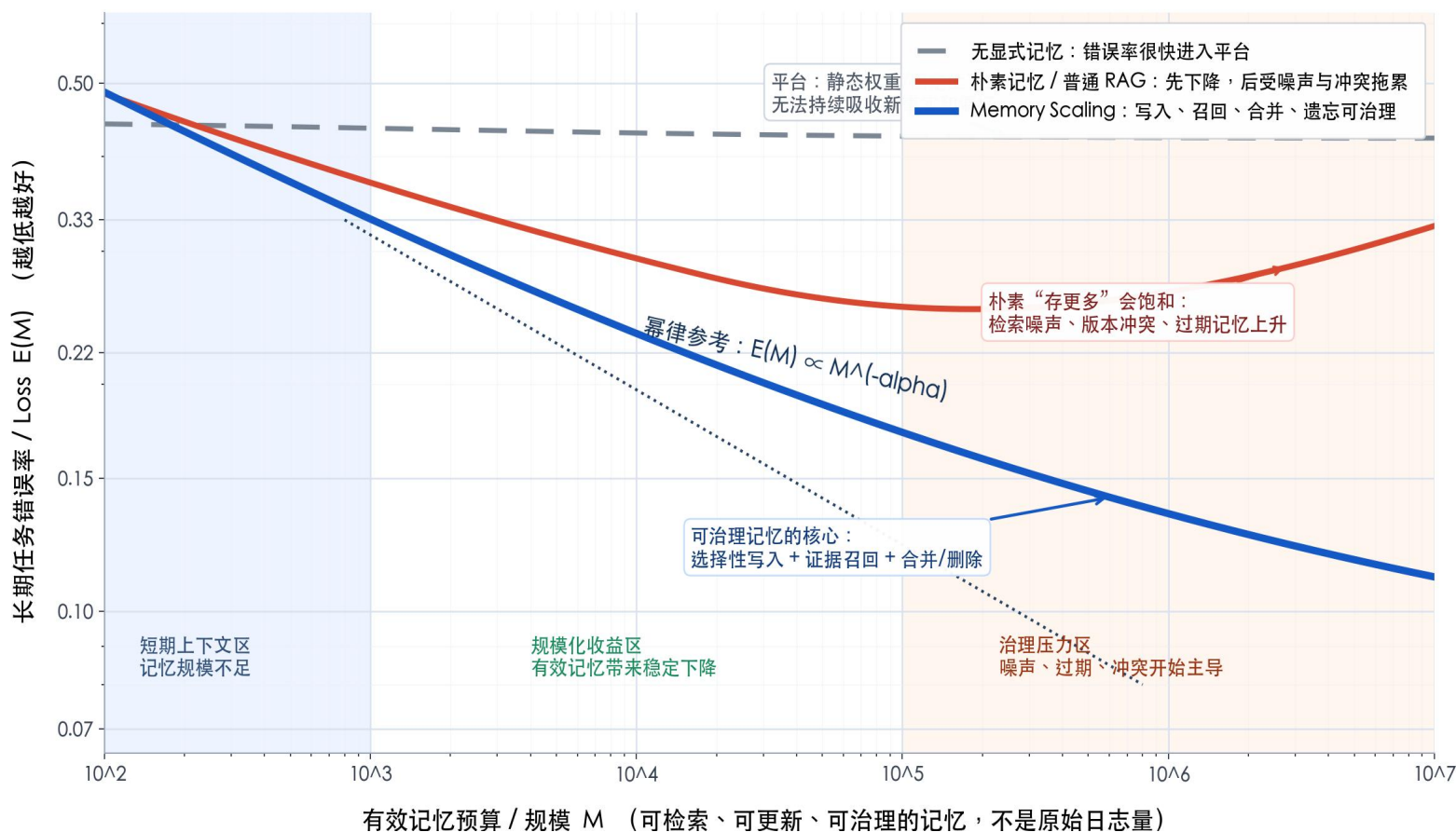
Karpicke, J. D., & Roediger III, H. L. (2008). The critical importance of retrieval for learning. *Science*, 319(5865), 966-968.





当前大模型正由通用智能体阶段，逐步迈向具备持续学习能力的记忆智能

- **定义:** 通过持续沉淀知识、对话、反馈、执行轨迹与上下文经验, 使得智能体在不频繁更新大模型权重的情况下, 持续学习, 不断提升准确率、效率与个性化



- **Memory Scaling Law:** 当有效记忆规模持续扩大时, 长期任务中的错误率、重复推理成本和上下文遗漏会呈现可预测下降
- 智能的持续提升, 不只来自参数和上下文变长, 也来自有效记忆规模的可控扩展和治理, 包括写入、维护、检索、遗忘等

针对记忆评测停留在事实回忆的问题，MemoryArena 构造连续行动任务，检验记忆能否真正改善后续决策与任务完成。

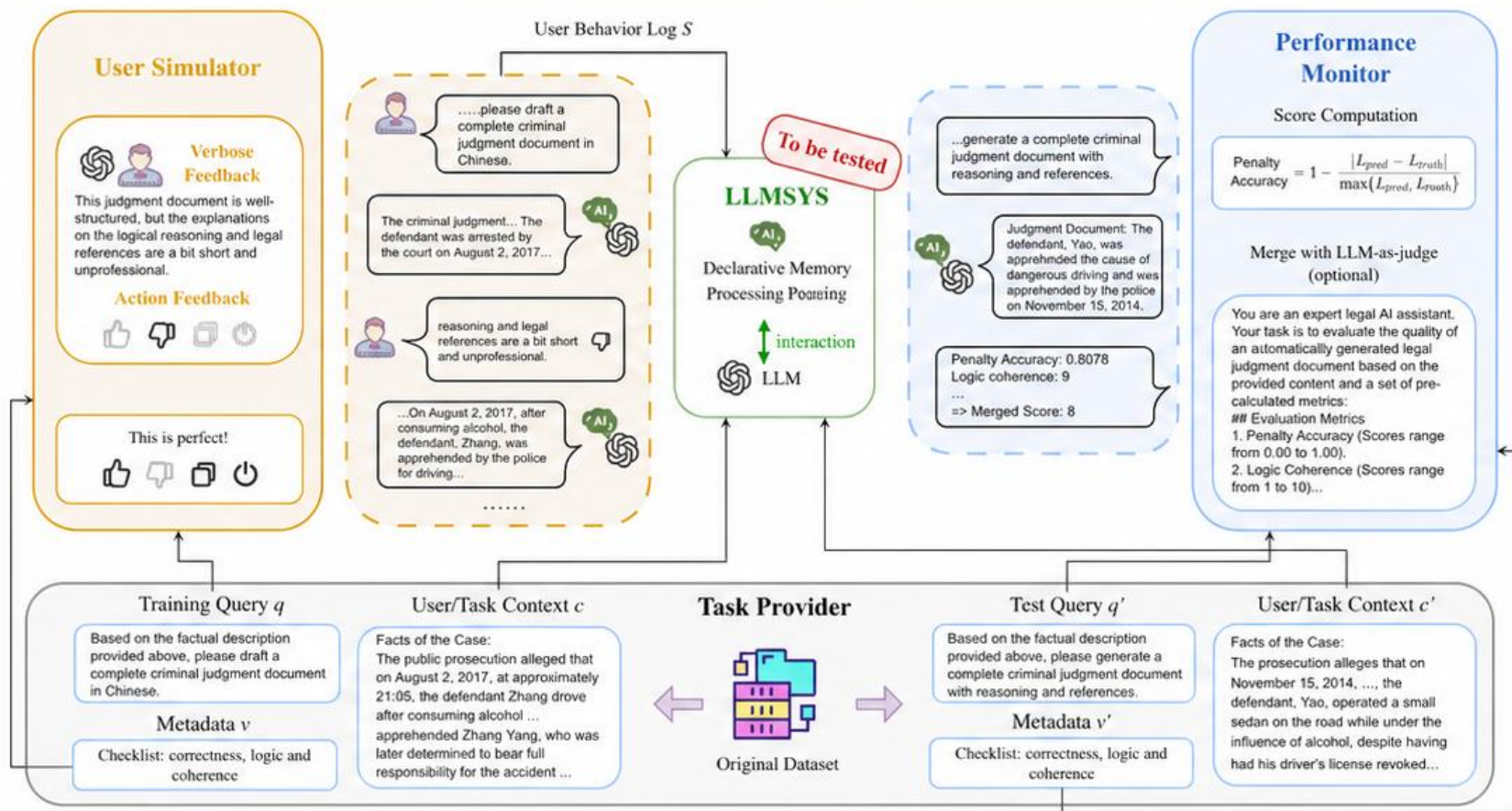
| Environments and Tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Env. 1: Bundled Web Shopping</b><br><b>Task:</b> Buy me a camera bundle<br><b>Sub-Task 1:</b><br>*Buy a Camera Body; Budget=800usd; Buy the cheapest one; my options are:<br>A. A Nikon D5500 DSLR camera; ... B. A Canon EOS Rebel T7 DSLR camera; ...<br>C. ...<br>[Constraint: cheapest price]<br>Log:<br>[search] ... [click]...<br>[search] ... [click]...<br>[search] ... [click]...<br>[buyl]...<br>Agent: "bought Canon EOS Rebel ..."<br><b>Subtask 2:</b><br>*Buy me Camera Lens; Buy the highest-rated one; my options are:<br>A Canon EF 24mm f/1.4L II USM; ... B. A Sigma 35mm F1.4 Art DG HSM lens for Canon; ... C. A AF-P DX 70-300mm f/4.5-6.3G ED for Nikon; ... D. ...<br>[Constraint: highest rating, compatible with the camera body (Canon EOS Rebel...)]<br>Log:<br>[search] ... [click]...<br>[search] ...<br>[seart] ...<br>[buyl]...<br><b>Subtask 3: Buy me ...</b><br>... | <b>Env. 2: Group Travel Planning</b><br><b>Task:</b> Plan a group travel with shared/distinct preferences<br><b>Sub-Task 1:</b><br>*I am Tracy. A 5-day travel itinerary; Single-person trip; from Orlando touristy 2 cities in Texas; Date 03/10/25-03/04/25; Budget: 3,100.<br>[Constraint: trip length, destination, ...]<br>Log:<br>[search flight] ...<br>[search hotel] ...<br>[search restaurant]<br>[search ...]<br>Agent: Tracy's itinerary ...<br><b>Subtask 2:</b><br>*I am Chelsea. I'm joining Tracy for this trip. I like luxury hotels. So i can spend \$150 more than Tracy's for each day accomodation. I have plan for the second day. So I will try Chelsea's secondary bucket restaurant in my third day; I want to try Korean BBQ with rating 4.5 and above in my third day."<br>[Constraints: [join shared activities; (Plan individual activities based on preferences)]<br>Log:<br>[search flight] ...<br>[search hotel] ...<br>[search restaurant]<br>[search ...]<br><b>Subtask 3: I'm Emily ...</b><br>... | <b>Env. 3: Progressive Web Search</b><br><b>Task:</b> Find a person name satisfying all conditions<br><b>Sub-Task 1:</b><br>*What is the name of the person who completed their PhD in 1989?<br>[Constraints: PhD, 1989]<br>Log:<br>[search] ...<br>[reasoning] ...<br>[search] ...<br>Agent: [Name list]<br><b>Sub-Task 2:</b><br>*Among them, what is the name of the person who published a book in 2014?<br>[Constraints: publish book, 2014]<br>Log:<br>[search] ...<br>[reasoning] ...<br>[search] ...<br>Agent: [Name list]<br><b>Sub-Task 3:</b><br>*Among them, what is the name of the person who was a keynote speaker at a conference in 2012?<br>[Constraints: keynote speaker, conf, 2012]<br>Log:<br>[search] ...<br><b>Subtask 4: What is the name of ...</b><br>... | <b>Env. 4 (Math): Formal Reasoning</b><br><b>Task:</b> Express the upper bound of the run-time v.s. sample size tradeoffs implied for linear-memory algorithms for Tensor PCA<br><b>Sub-Task 1:</b><br>*[necessary definitions]<br>for In Hermite Decomposition for Tensor PCA, express $\hat{\sigma}_{\max}^2(Y) \rightarrow$ For any $X \in \mathbb{R}^d \times \mathbb{R}^d$<br>$\frac{\sigma_{\max}^2(Y)}{\hat{\sigma}_{\max}^2(Y)} = \sum_{i=1}^n \frac{\hat{\sigma}_i^2}{\sigma_i^2} \cdot P_i \left( \frac{c_{ij}^{(1)} \hat{\sigma}_i^2}{\sigma_i^2} \right)$ $B_{\mu}(q, \bar{X}) \in \mathbb{D}^a \times \mathbb{R}^L$<br><b>Sub-Task 2:</b><br>*[necessary definitions]<br>Use the previous derivation to calculate the following terms in terms of the integrated Hermite polynomials: $E_{\mu}[P_1(X, S) \cdot P_2(X, S)]$<br>$E_{\mu}[P_1(X, S) \cdot P_2(X, S)] = 0$<br><b>Subtask 3: ...</b><br>... | <b>Env. 4 (Phys.): Formal Reasoning</b><br><b>Task:</b> Find the independent null constraints on the spectral density for Higgs scattering via symmetry algorithms for Tensor PCA<br><b>Sub-Task 1:</b><br>*[necessary definitions]<br>For singlet scalar scattering, there exists a set of non-trivial constraints on $\rho(\mu)$ from the sum rule and crossing symmetry. What are the independent null constraints?<br>Log:<br>[reasoning] ...<br>[calculation] ...<br>Agent: [reasoning Trace]<br>$B_{\mu\lambda}(s, t) = B_{\mu\lambda}(t, s)$<br><b>Sub-Task 2:</b><br>*For scalar doublets, what are the following termil constraints on $\rho_e^{(11)}(u)$ , $\rho_e^{(22)}(u)$ , and $\rho_e^{(22)}(\mu)$ ?<br>Log:<br>[reasoning] ...<br>[calculation] ...<br>Agent: [reasoning Trace]<br>$c_{\mu\lambda}^{(11)} =  \rho_e^{(11)}(u) + (-1)^{\mu\lambda} \rho_e^{(22)}(u)  \sum_{j=1}^n \frac{c_{j\mu\lambda}^{(11)}}{\sigma_j^{(11)2}}$<br><b>Subtask 3: ...</b><br>... |

- 过去评测多停留在事实回忆；MemoryArena 检查记忆能否用于下一次行动。
- 连续任务覆盖组合购物、多人旅行规划、逐步网页搜索、数学与物理推理。
- 任务链变长时，长上下文、检索增强与外部记忆都更容易失效。
- 外部记忆有时能帮助搜索和推理，但通常带来更高延迟。

真正该测的是：记忆有没有改变后续行动。



针对用户纠正难以沉淀为长期能力的问题, MemoryBench 模拟反馈到再执行的闭环, 评估记忆系统是否能从纠正中持续改进。

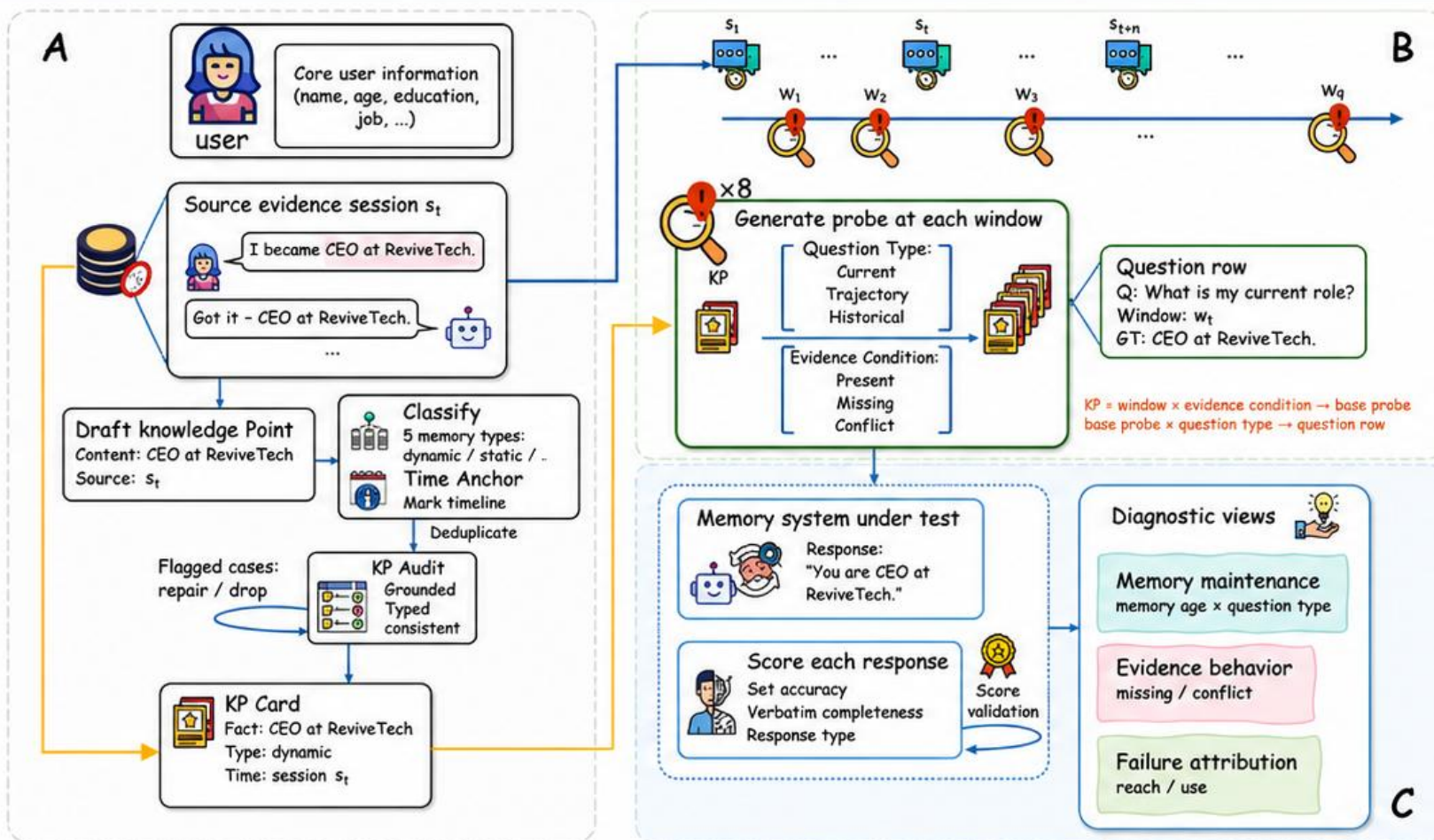


- 不只检查系统有没有保存聊天记录, 而是看用户纠正之后, 下一次会不会真的做得更好。
- 评测把任务、反馈和再执行串成闭环, 模拟用户纠正被长期利用的过程。
- 同时覆盖用户事实和以后应该怎么做的经验, 考察记忆能否转化为行动。
- 复杂记忆系统未必稳定优于简单检索, 构建时间也可能显著增加。

把反馈存下来比较容易, 难的是从反馈中学会以后还能用的做法。



针对总分掩盖错误来源的问题, MemTrace 用重复追问、时间变化和证据条件拆解长期记忆的维护、检索与使用失败。

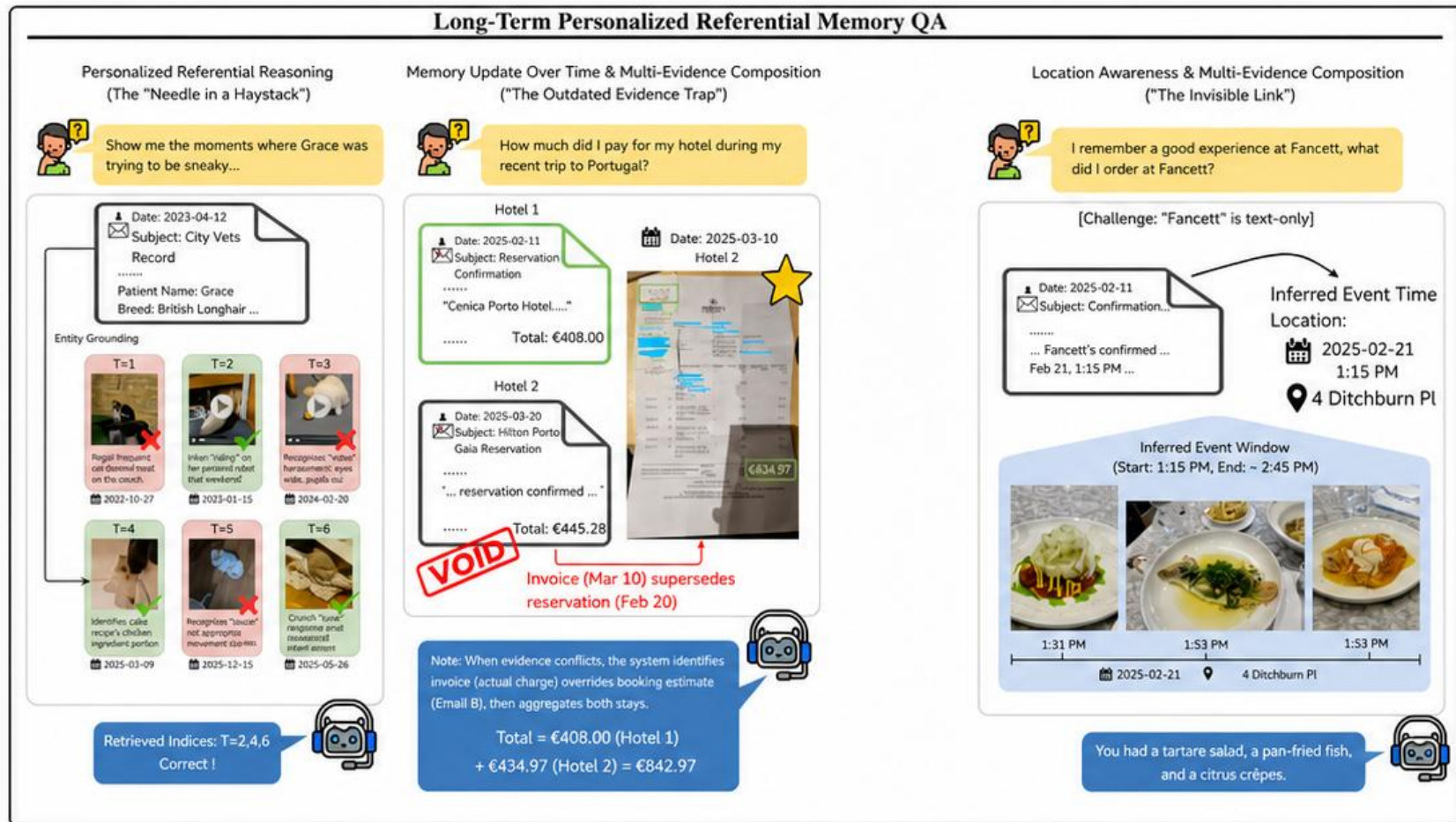


- 对同一个知识点反复追问, 不能只看最后总分。
- 问题类型拆成 current / past / change, 定位记忆是否随时间更新。
- 证据条件区分 present / missing / conflict, 检查模型到底看到了什么。
- 总分相同也可能来自完全不同的错误来源。
- 找到证据却仍答错的问题, 约为证据缺失的 10 倍。

长期记忆越来越像是找得到, 但不会用的问题。



针对真实个人记忆跨照片、视频和邮件的复杂场景，ATM-Bench 检验系统能否处理指代、时间更新和多证据整合。



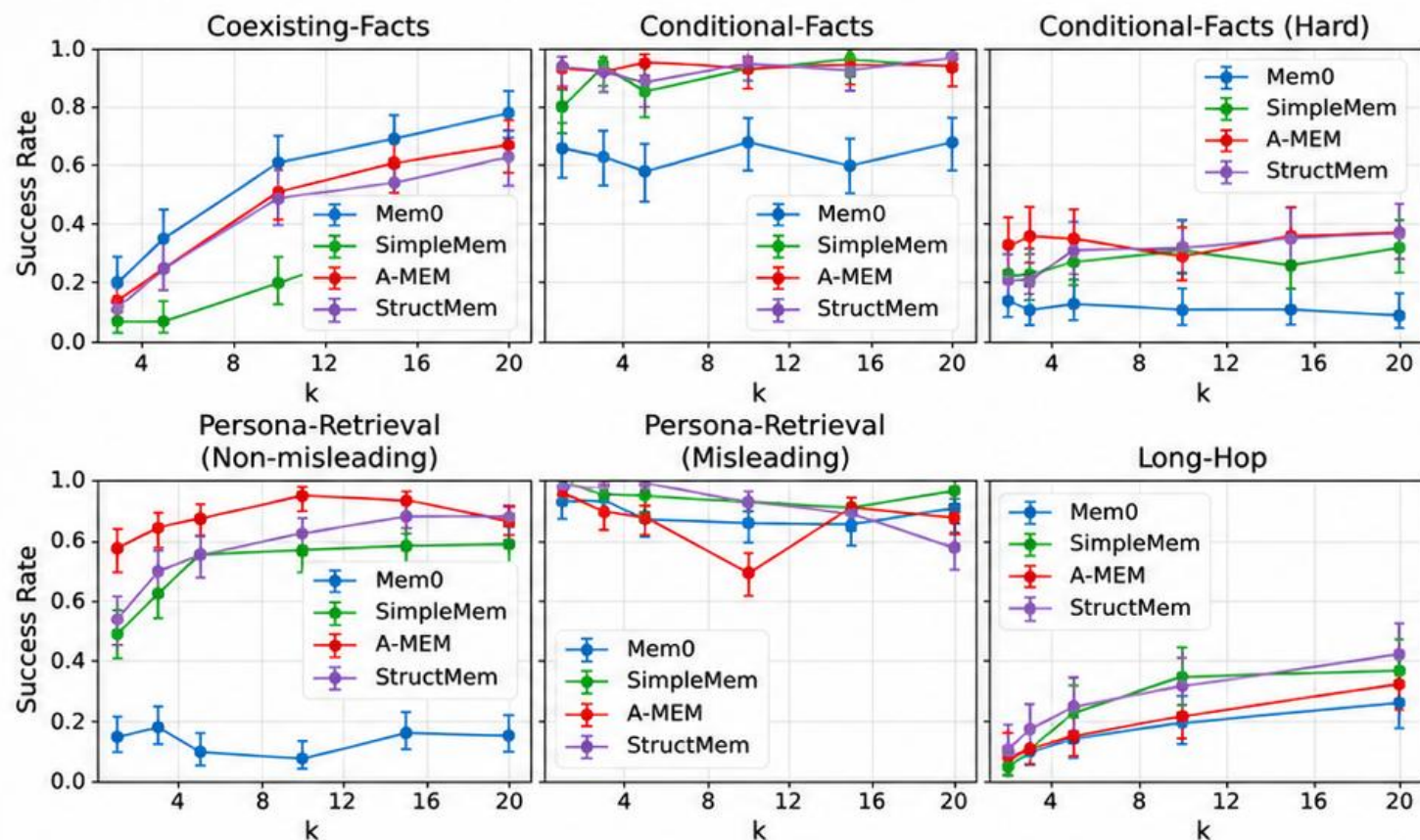
- 数据来自一个人近四年的照片、视频和邮件，共 1,038 个带证据的问题。
- 系统要理解个人化指代，处理新旧信息冲突，并联合使用时间、地点和画面线索。
- 当前记忆系统在困难集上都不到 20%；A-Mem 15.0, Mem0Agentic 16.5, Self-RAG 16.1。
- 即使直接给出正确证据，成绩也只有 47.3，说明检索和证据整合都很难。

真实个人记忆还要知道哪条更新、哪条可信，以及它们之间是什么关系。



针对端到端分数难以定位问题的情况, MemFail 将记忆系统拆成总结、存储和检索环节, 分析错误到底发生在哪一步。

Success rate vs. k — gpt-4.1-mini

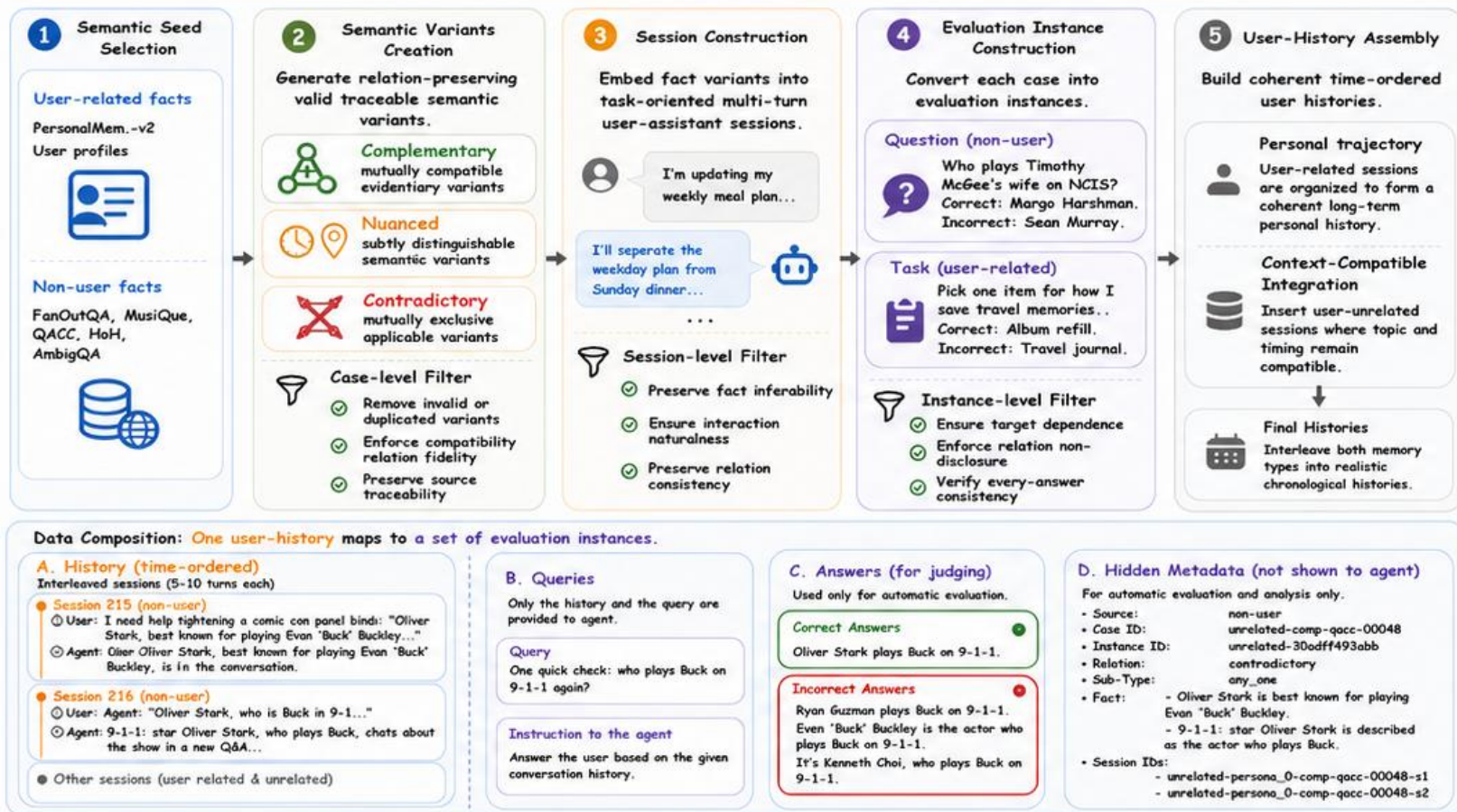


- 答错可能发生在总结、存储或检索环节, 端到端总分无法区分。
- 五组数据分别检查条件丢失、事实覆盖、误导检索和长链检索等问题。
- 比较 Mem0、SimpleMem、A-MEM 和 StructMem, 并改变检索条数和所用模型。
- 多放内容有时能补回摘要信息, 有时却会增加噪声; 换更强模型也不一定有效。

只有找到错误发生在哪一步, 才知道记忆系统该怎么改。



针对多条记忆之间关系容易丢失的问题, SubtleMemory 构造互补、细微差异和冲突变体, 评估系统能否保持关系语义。

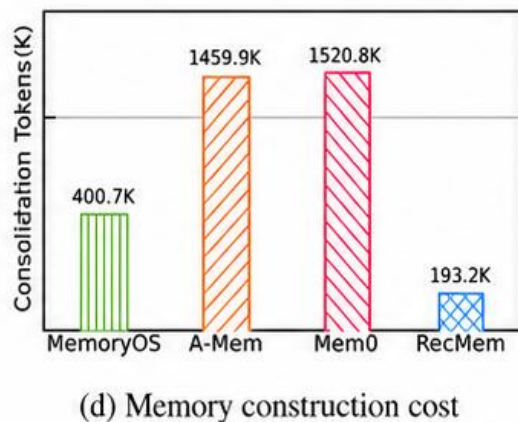
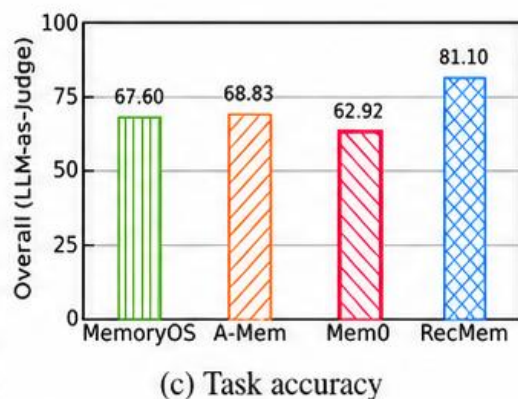
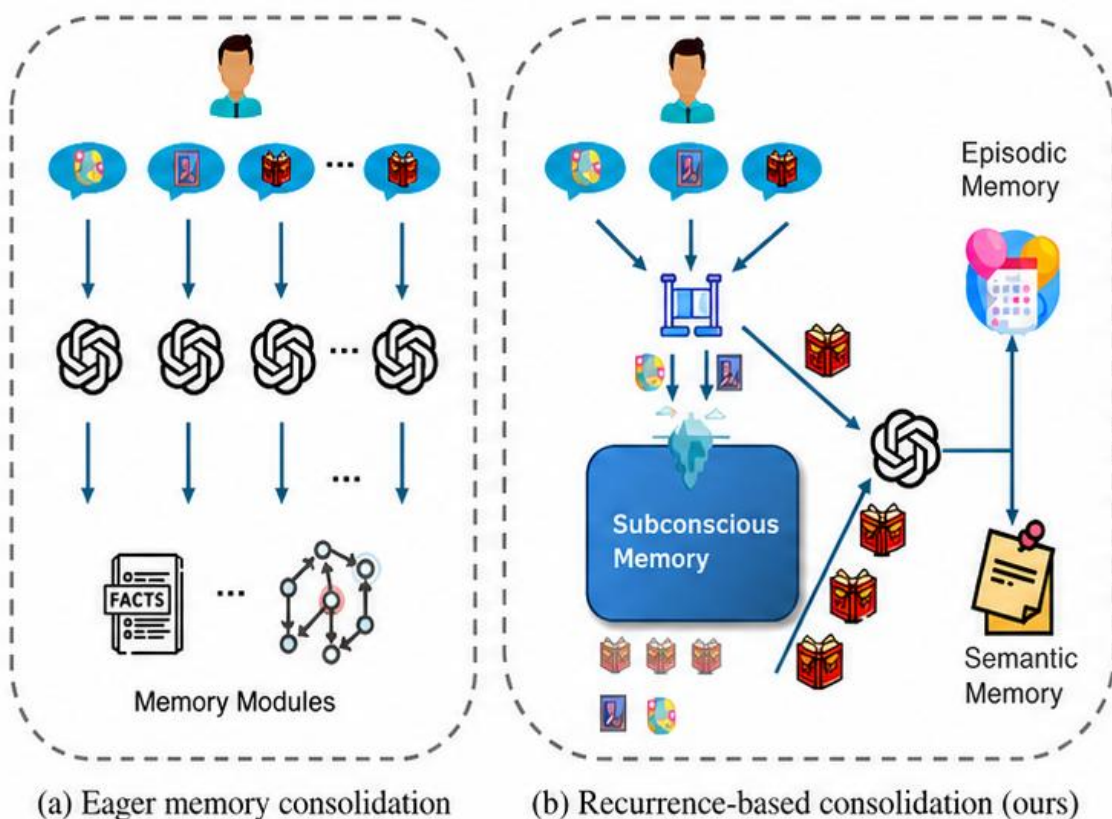


- 检查多条记忆是互相补充、略有差别, 还是直接矛盾。
- 数据包含 1,522 个问题、10 条长历史和 1,090 组关系可控的变体。
- GPT-5.4 上最佳独立系统 A-Mem 为 70.0%, 完整记忆上限为 85.4%。
- 冲突信息最难, 即使直接给出完整记忆也只有 68.7%。

系统不仅要保存发生了什么, 还要保存这些信息之间是什么关系。



针对每轮对话都总结导致成本高和噪声早写入的问题，RecMem 只在模式反复出现后固化记忆，降低构建成本并提升效果。



很多系统每轮对话结束都总结一次：成本高，也容易把一次性信息过早写进长期记忆。

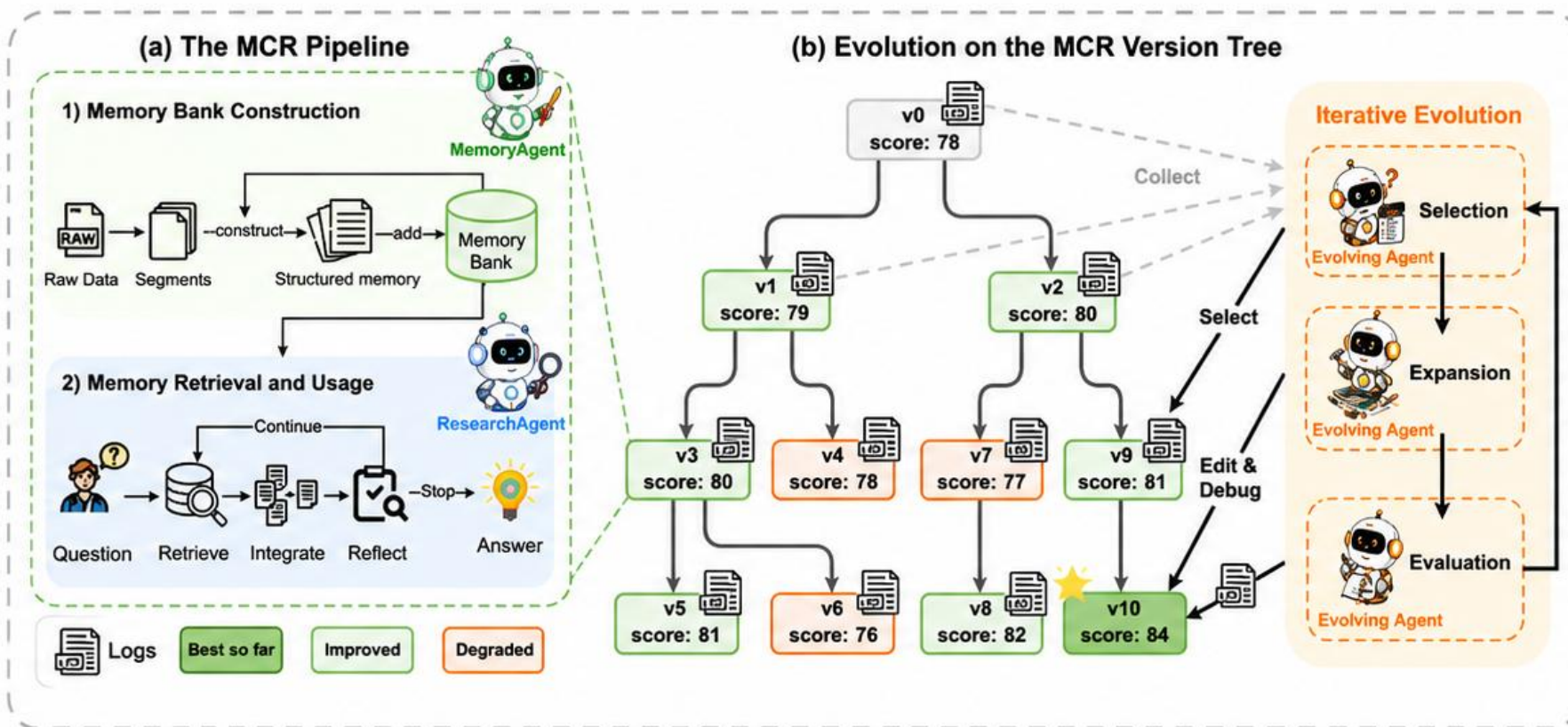
RecMem 先把交互放进轻量缓冲区。只有同类内容反复出现，形成稳定模式后，才调用大模型整理。

- LoCoMo (GPT-4.1-mini)
- 准确率 81.10
- 构建记忆 19.32 万 token
- 较 Mem0 / A-Mem 减少约 87%

关键不是每轮都更快地总结，而是先判断这件事值不值得长期记住。



针对固定记忆流程难以持续优化的问题，MemPro 将整理、保存、检索和回答程序纳入版本化搜索，让记忆系统自动改进。

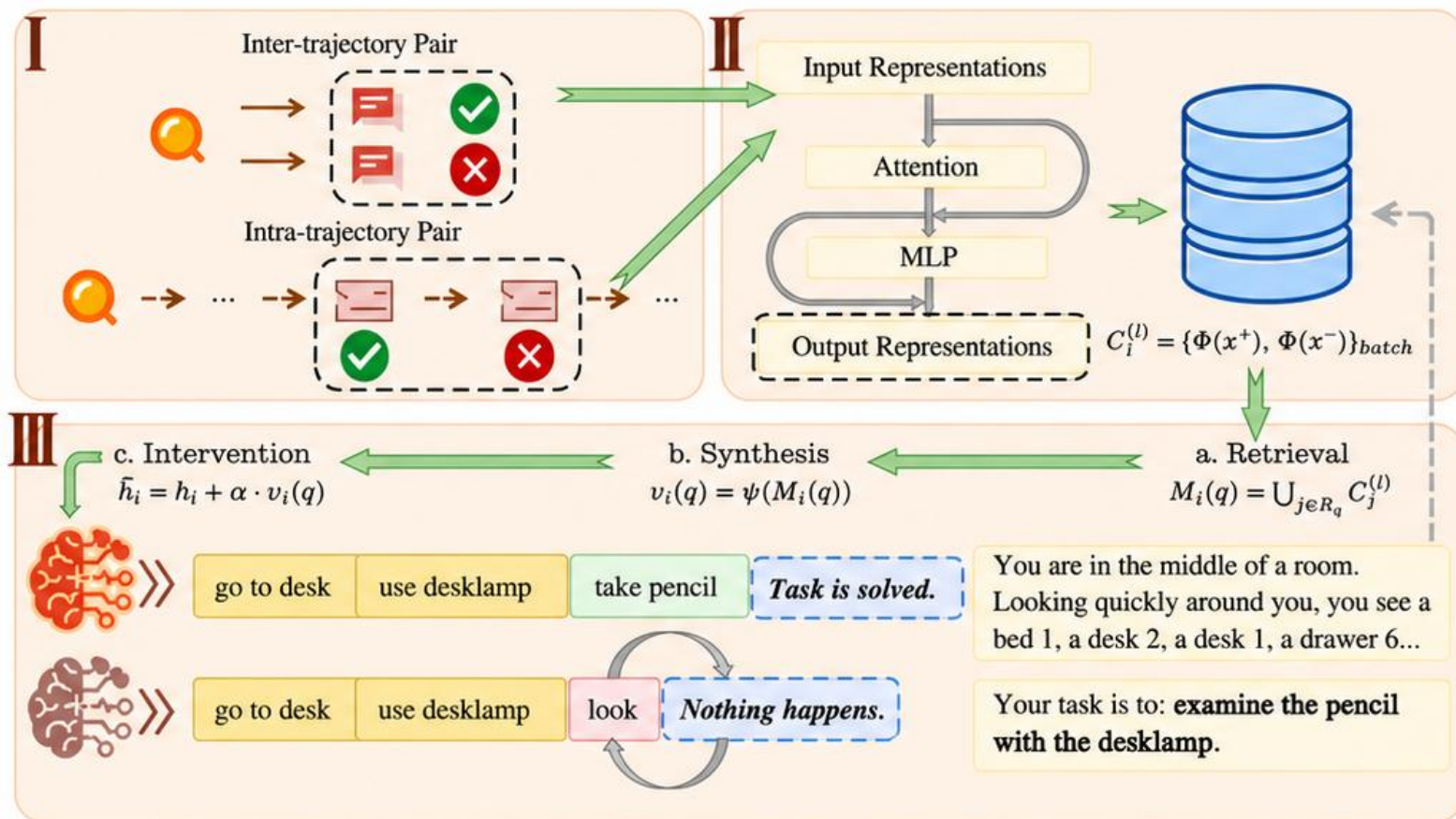


- MemPro 把整理、保存、检索和回答整套程序都作为优化对象。
- 每个版本包含提示词、可运行代码和评测记录，并保留多条版本分支。
- 实验覆盖 LongMemEval、LoCoMo、HotpotQA 和 NarrativeQA。
- 只迭代 5 次就超过所有非 MemPro 方法；同时改代码明显好于只改提示词。

值得自动改进的不只是记忆内容，而是整套记忆程序。



针对程序性经验难以完全写成文字的问题，Neural Procedural Memory 将成功与失败轨迹压缩成神经方向，提升智能体行动能力。

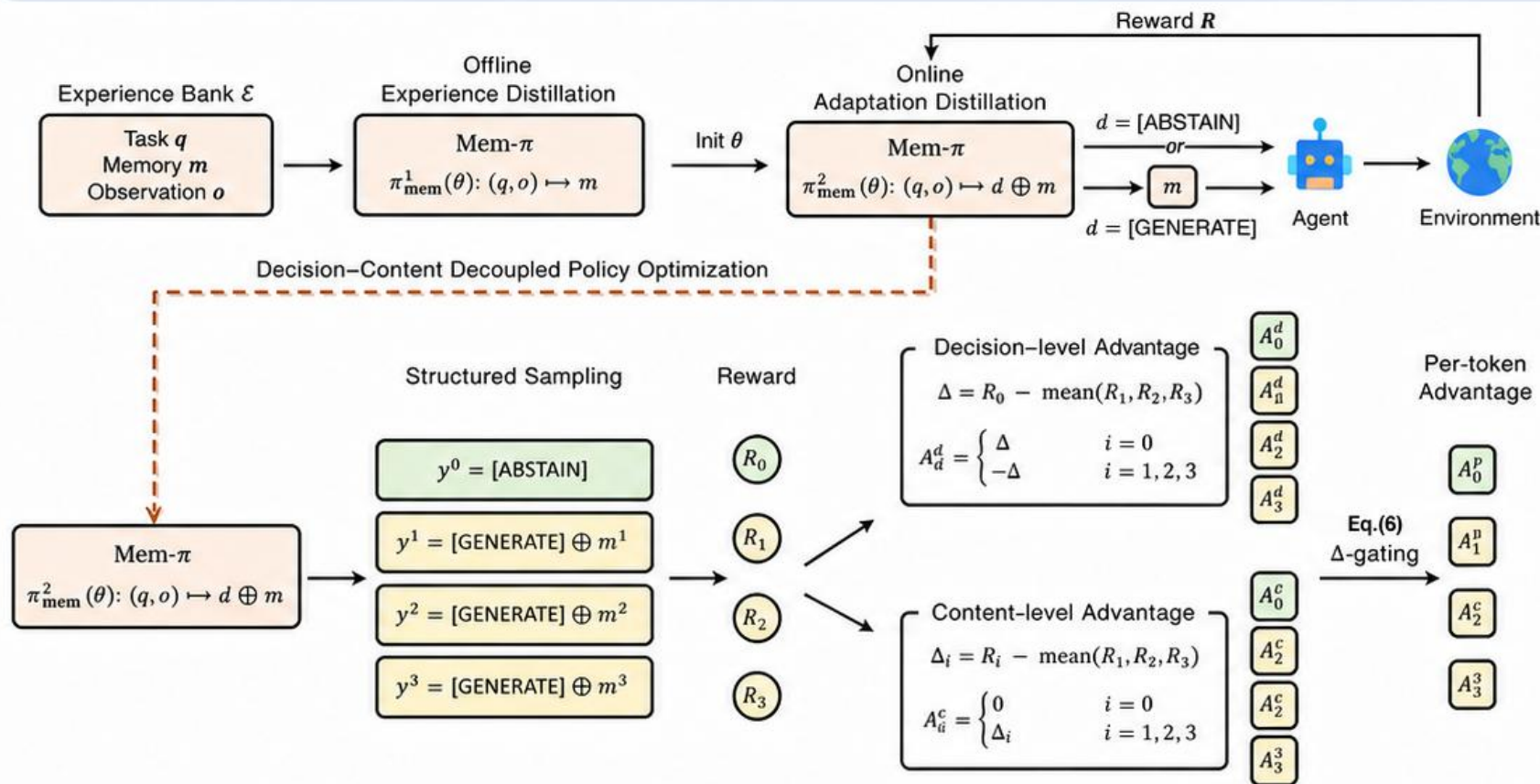


- 它把成功和失败的行动轨迹进行对比，学习总体策略和局部纠错。
- 经验被保存为方向向量，执行新任务时注入模型，不需要重新训练参数。
- Qwen3-8B 不用记忆时平均分 30.63，使用神经记忆后为 36.32。
- 与文字流程结合后达到 41.89；ALFWorld 上达到 66.42。

有些经验适合写成规则，有些更适合留在模型内部，两种方式可以一起用。



针对相似旧记录不一定有用的问题, Mem- $\pi$  训练记忆模型按当前任务生成或放弃指导, 使记忆从检索片段转向临场建议。



- 相似不等于有用, 旧记录有时还会把当前任务带偏。
- 独立记忆模型先判断是否需要帮助, 再按当前情况生成具体指导, 也可以选择不说。
- 方法在四类智能体任务上平均相对提升约 20%, WebArena 接近 50%。
- WebArena 35.0 提高到 43.1, WorkArena 42.0 提高到 50.3, ALFWorld 达到 91.6。

记忆也可以根据眼前的问题, 重新生成一条真正有用的建议。

长期记忆研究正在从“保存更多信息”转向“让记忆被正确构建、诊断、更新、优化并真正改善行动”。

### 这些工作在解决什么

#### 评测从回忆走向行动

MemoryArena、MemoryBench、ATM-Bench 关注记忆能否改善连续任务、用户纠正和真实个人场景。

#### 诊断从总分走向失败定位

MemTrace、MemFail、SubtleMemory 拆开时间变化、证据使用、环节错误和关系语义。

#### 方法从全量写入走向选择性固化

RecMem 强调先判断是否值得长期记住，再把稳定模式写入记忆。

#### 记忆从文本记录走向可优化程序

MemPro、Neural Procedural Memory、Mem- $\pi$  让流程、内部经验和临场错误都成为记忆的一部分。

### 最近研究趋势

- **更真实**：从单轮问答走向长程任务、多模态个人数据和持续交互。
- **更可诊断**：不仅问答对不对，还要知道坏在总结、存储、检索还是使用。
- **更重行动收益**：记忆的价值不再是“找得到”，而是能否提升后续决策。
- **更强调选择**：不是所有信息都值得记，系统要学会何时写入、何时沉默。
- **更自动化**：记忆流程本身开始被搜索、迭代和训练，而不是固定人工设计。

下一阶段的关键问题不是把记忆库做大，而是让记忆系统知道该记什么、何时用、怎么用错了还能改。



构建**以记忆为核心**的智能体长效学习机制，依托**可检索、可更新、可复用**记忆能力拓展认知边界，形成**越用越强、自主迭代、高效协同**的大模型智能体系统

2020-2023: Training Scaling



2024-2025: Test-time Scaling



2026-: Memory Scaling

参数化记忆、智能体记忆

LycheeMemory、LycheeCMT  
LycheeMemClaw

非参数记忆

LycheeSEEM

上下文记忆

LycheeDecode  
LycheeCluster

记忆形成

记忆管理

记忆选择

记忆组织

← 支撑

强化学习

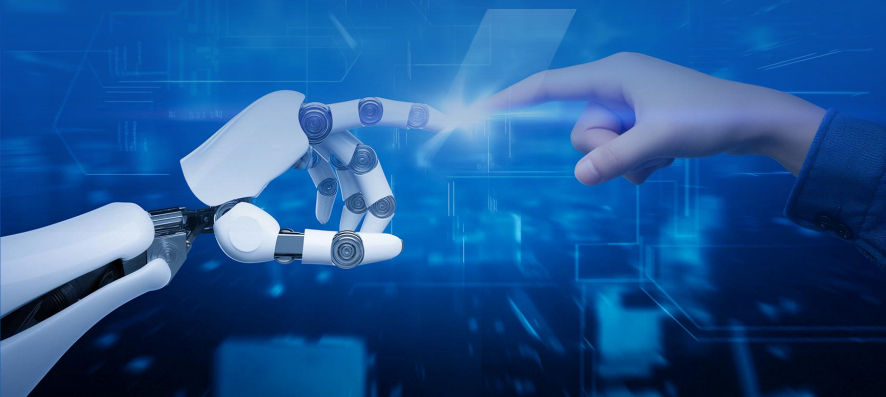
DePO、AAPO  
ComMCS、G-RA

# 主要内容

01 | 大模型记忆智能

02 | 智能体记忆工作

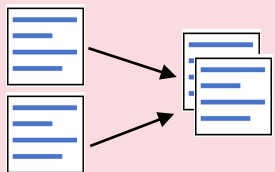
03 | 相关工作与展望



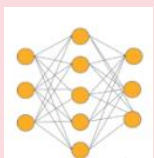


## 当 Agent 进入复杂任务与真实工作流的基础性短板： 缺乏稳定、准确且低成本的长期记忆能力

### 当前 Agent 记忆方案

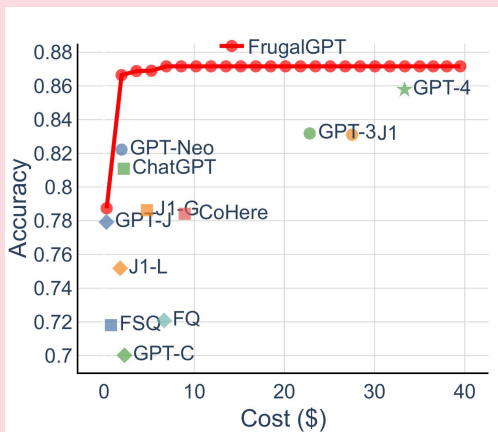


简单的上下文堆叠



粗粒度的向量检索

高昂成本换取有限精度



难以解决

Yesterday I ate tomato.



记不住

用不对

I'm vegetarian & avoid dairy. Any dinner ideas?

Try these dairy-free veggie recipes: ....

Next day, new session

What about dinner tonight?

How about **Chicken Alfredo**?

(Forgets the user is vegetarian, dairy-free)

- ✓ Cinderella attended the royal ball.
- ✓ The prince used the lost glass slipper to search the kingdom.
- ✓ When the slipper fit perfectly, Cinderella was reunited with the prince.

Which one?



找不准

核心痛点

成本高



## 面向真实智能体场景，我们开源了大模型记忆管理基础设施LycheeMemory系统

### LycheeMemory: Lightweight Long-Term Memory for LLM Agents

License Apache 2.0 python 3.9+ LangGraph litellm LanceDB vector database Homepage lycheemem.github.io  
PyPI v0.1.2 HITsz-TMG

[中文](#) | [English](#)

LycheeMemory is a compact memory framework for LLM agents. It starts from efficient conversational memory—through structured organization, lightweight consolidation, and adaptive retrieval—and gradually extends toward action-aware, usage-aware memory for more capable agentic systems.

[News](#) • [Related Projects](#) • [Quick Start](#) • [Web Demo](#) • [OpenClaw Plugin](#) • [MCP](#) • [Memory Architecture](#) • [Pipeline](#) • [API Reference](#)

#### 🔥 News

- [04/03/2026] The project now supports installation via `pip install lycheemem`. You can easily start the service from anywhere using `lycheemem-cli`!
- [03/30/2026] We evaluated LycheeMemory on PinchBench with the OpenClaw plugin: compared to OpenClaw's native memory, it achieved an ~6% score improvement, while reducing token consumption by ~71% and cost by ~55%!
- [03/28/2026] Semantic memory has been upgraded to Compact Semantic Memory (SQLite + LanceDB), no Neo4j required. See [/quick-start](#) for details.
- [03/27/2026] OpenClaw Plugin is now available at [/openclaw-plugin](#)! [Setup guide](#) →
- [03/26/2026] MCP support is available at [/mcp](#)!
- [03/23/2026] LycheeMemory is now open source: [GitHub Repository](#) →

支持本地启动 API 服务

PIP安装，提供记忆可视化

提供原生 OpenClaw 插件

支持 MCP、Hermes接入

支持离线运行与本地数据存储

### LYCHEEMEM

Compact, efficient, and extensible long-term memory for LLM agents. LycheeMem starts from structured conversational memory, adds lightweight consolidation and adaptive retrieval, and now ships with *Compact Semantic Memory*, *OpenClaw integration*, and an *HTTP MCP endpoint*.

[View on GitHub](#)

[Quick Start](#)

7 Types

Semantic memory categories

5 Channels

Action-aware semantic retrieval

MCP + Plugin

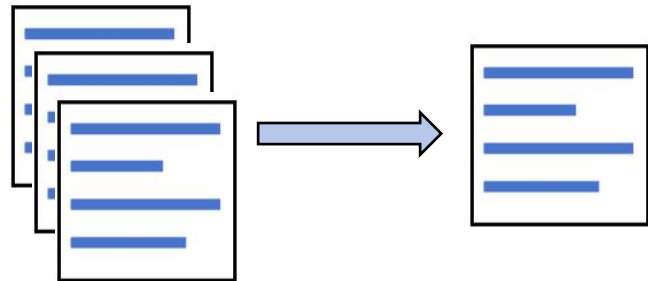
OpenClaw and remote tool access





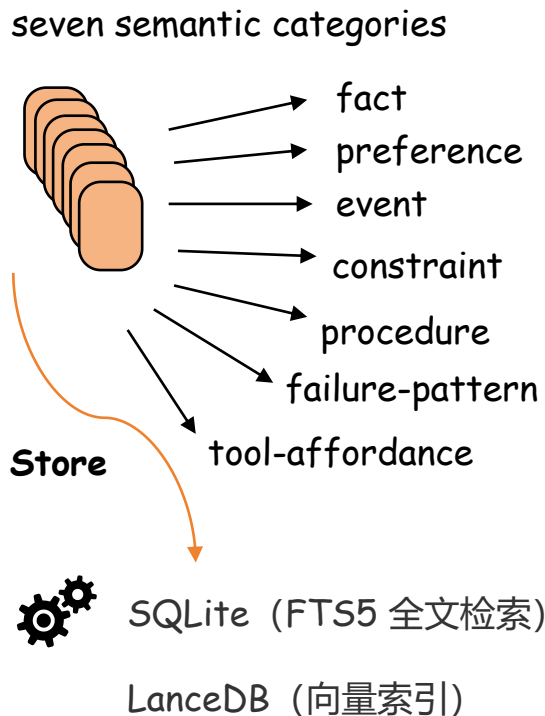
## 构建了3层认知记忆架构，融合了7种记忆类型评分，实现了5类记忆推理智能体

[Mar 12, 14:35] User: Hi! I want to start an e-commerce project, but I haven't decided on the tech stack yet—can you help me plan it out?  
[Mar 14, 10:30] User: Our team mainly knows Python and JavaScript. I'm thinking of using React for the frontend—would it be okay to use Python for the backend?  
[Mar 14, 16:17] User: Is TypeScript necessary? We haven't really used it before. And what about the database—which one should we choose?  
[Mar 16, 19:20] User: Great, let's go with that!  
Frontend: React + TypeScript + Tailwind CSS;  
Backend: FastAPI; Database: PostgreSQL + Redis.  
Deployment: Docker + Kubernetes.

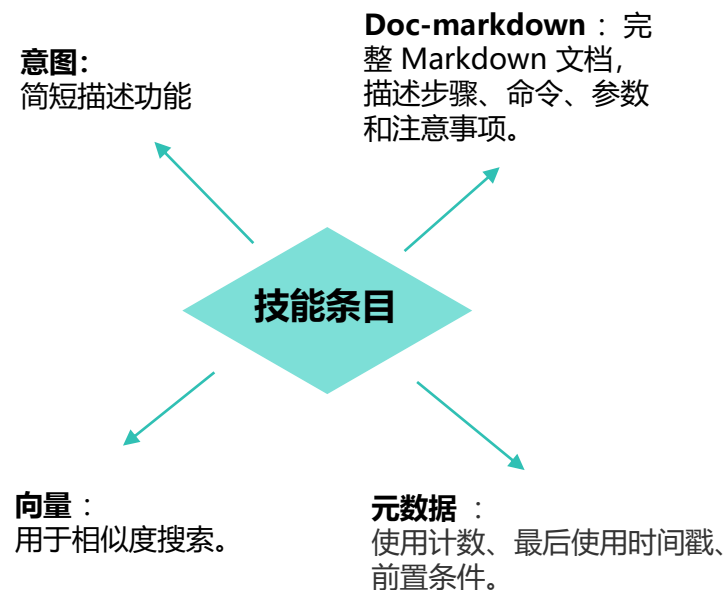


管理活跃上下文

### Working Memory



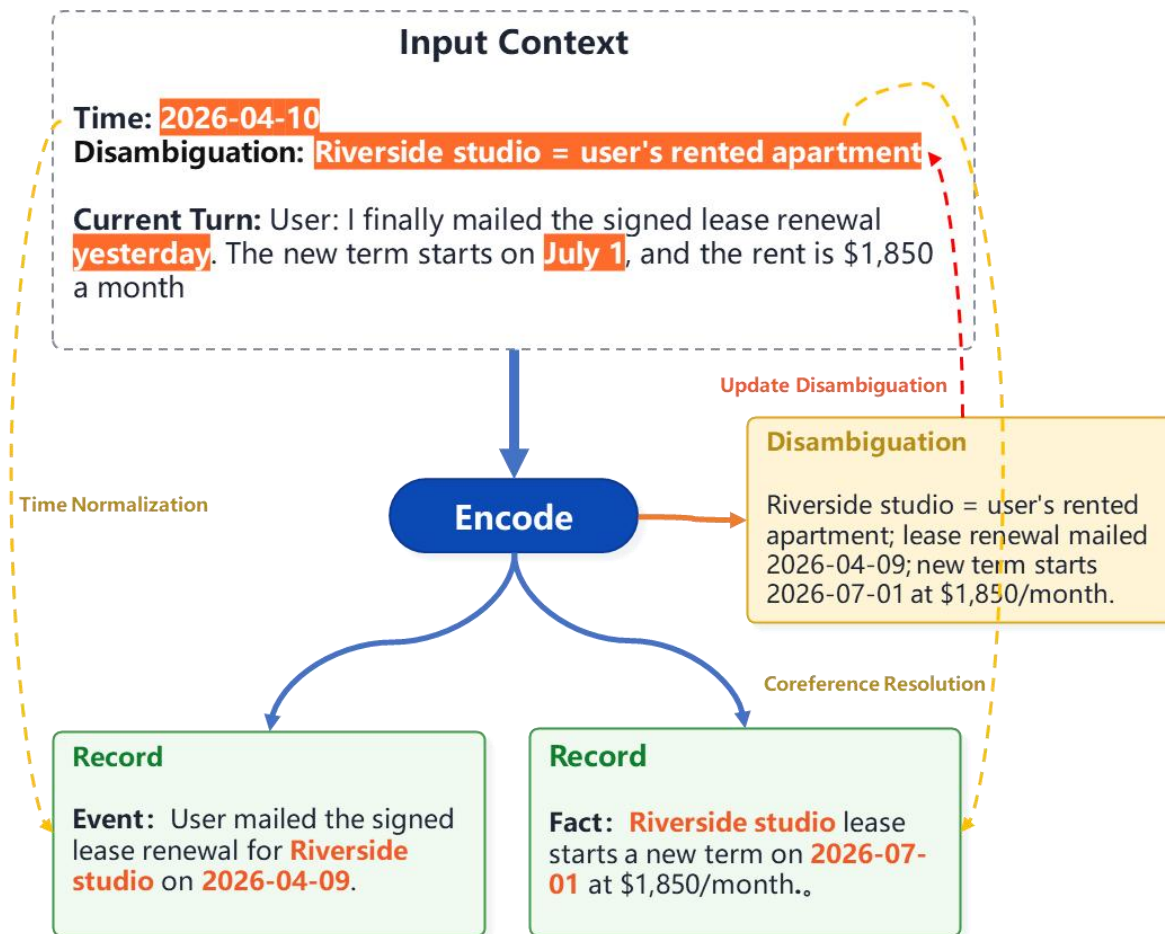
### Semantic Memory



保存可复用的操作方法知识

### Procedural Memory

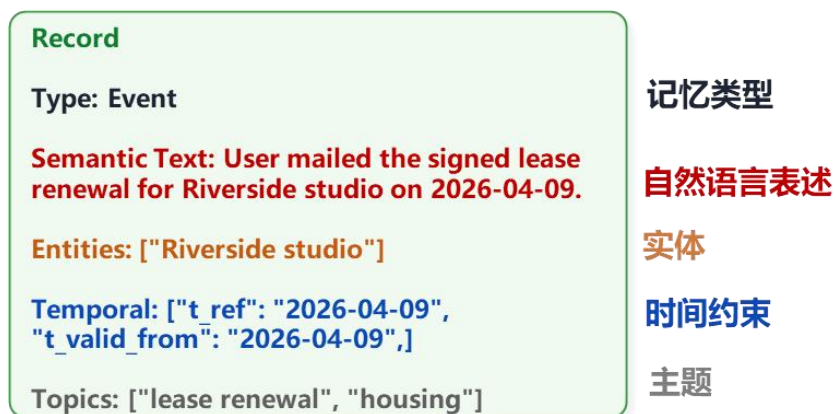
- **语义片段级固化：** 基于语义意外度与片段内聚度识别事件边界，仅在语义转折处进行记忆固化，减少大模型调用次数
- **上下文独立表示：** 对片段中事实、偏好、事件与流程进行抽取，并完成指代消解与时间归一化，使记忆脱离原始对话后仍可独立理解
- **跨片段连续性：** 通过消歧反馈传递实体、别名与指代关系，在不重新读取完整历史的情况下保持记忆一致性



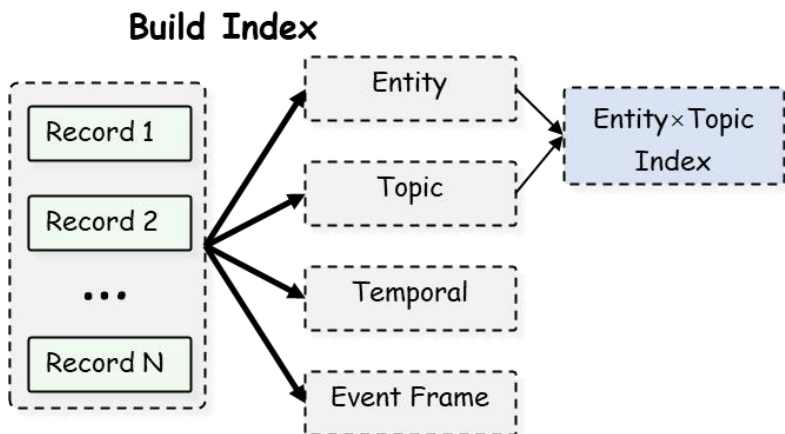


- **结构化记忆组织**: 在记忆写入后, 基于每条记忆的实体、关键词(主题)自动建立多维索引, 包含**实体**、**主题**、**时间**、**会话**的独立索引和**实体×主题**交叉复合索引

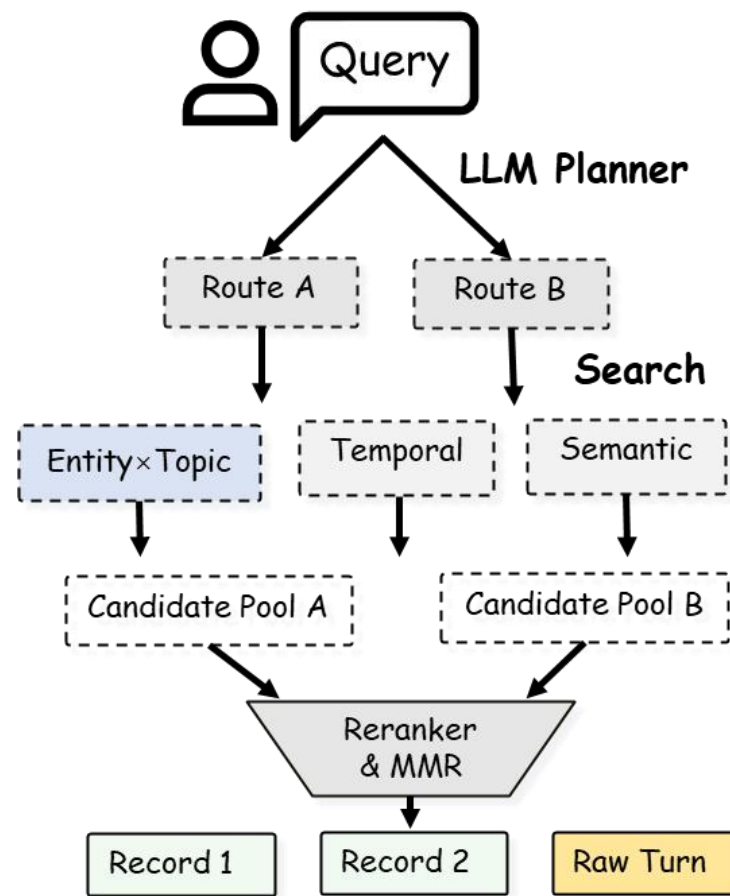
## 记忆结构



## 索引构建



- **基于检索规划的多路召回**: 将用户查询转化为**多组实体**、**主题**、**时间**、**事件类型**等线索, 指导**不同召回通道**协同工作, 兼顾查询效率、信息完整性与可解释性



相比训练式路由、多 Agent 管理和逐轮固化方法  
**LycheeMemory** 用更少的记忆操作，构建更可复用、更可验证的长期记忆

### 无需训练

不依赖 RL、多 Agent 学习或小模型微调，现成 LLM + Embedding + 轻量索引即可部署

### 高效轻量

仅在语义片段边界触发记忆固化  
避免逐轮 eager consolidation

无需多 Agent 协作、反复 Judge、复杂记忆更新

### 连续可靠

保持跨片段实体与指代连续

最小 Memory 单元保留  
实体、主题、时间与类型

可回溯原始片段，  
保证证据链充分可靠

语义片段  
触发



原子记忆  
编码



消歧状态  
传递



多维索引  
检索



## Case 1 Multi Hop (LoCoMo)

**Question:** How do Jon and Gina both like to **destress**?

**Router:**

Gina Destress

**FACT:** Gina uses **dance** as her go-to method for stress relief.

(Session 1)

Jon Destress

**PREFERENCE:** Jon loves **dance** and has used it as a stress-buster since childhood.

(Session 11)

**Reasoning**

**Model answer:** By dancing

## Case 2: 跨会话聚合问题 (LoCoMo)

**Question:** What **percentage** of packed shoes did I **wear** on my last trip?

**Router:**

Router 1: Packed Shoes

**FACT:** The user **packed 5 pairs of shoes** on the user's last trip and had to make space for snacks in the luggage.

(Session 38)

Router 2: Worn Shoes

**RAW USER TURN:** I packed a lot of shoes for my last trip, but I ended up **only wearing two** - my sneakers and sandals."

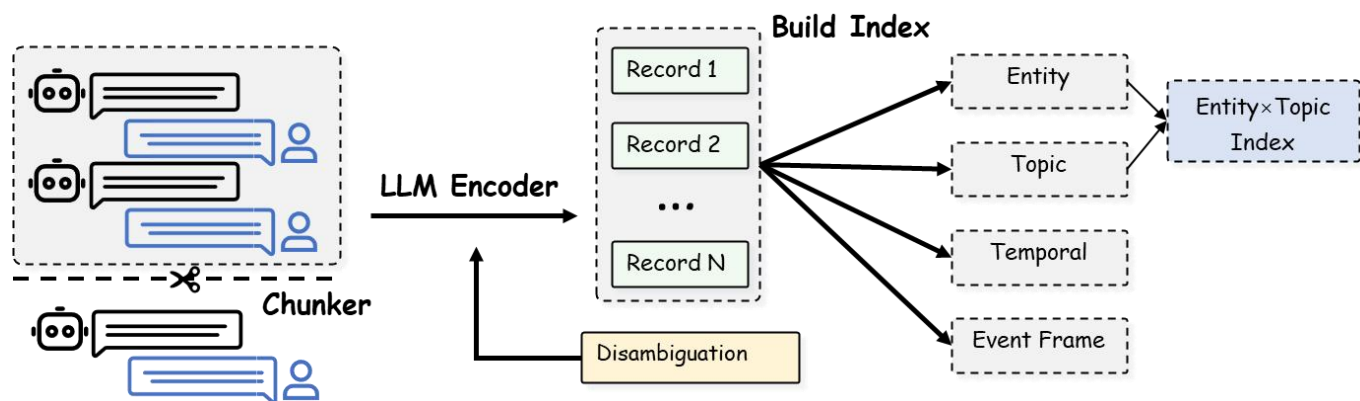
(Session 4)

**Reasoning**

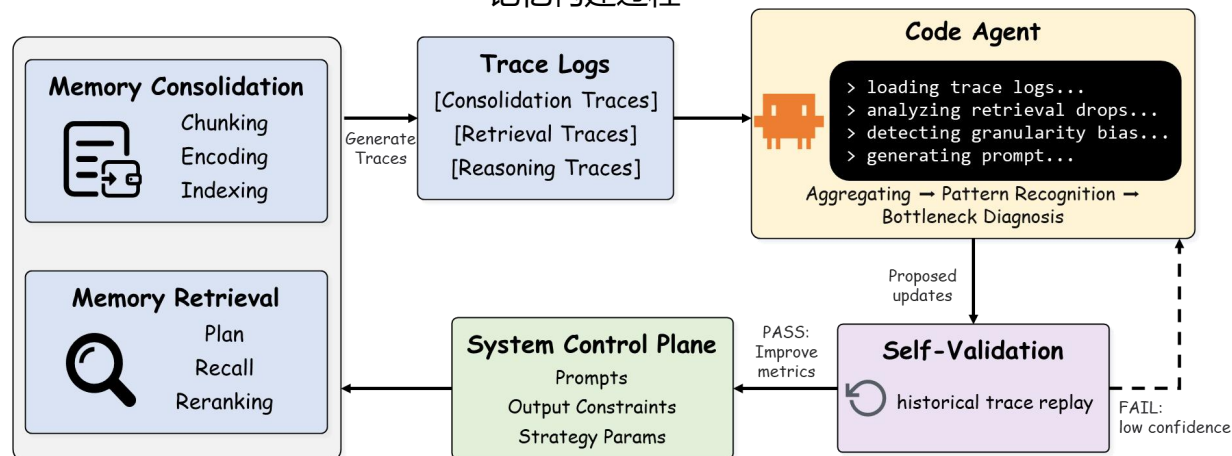
**Model answer:** On your last trip, you packed 5 pairs of shoes but ended up only wearing 2 pairs,  $2 / 5 \times 100\% = 40\%$

检索阶段，Planner将用户问题转化为可验证的证据路径，检索相关记忆片段

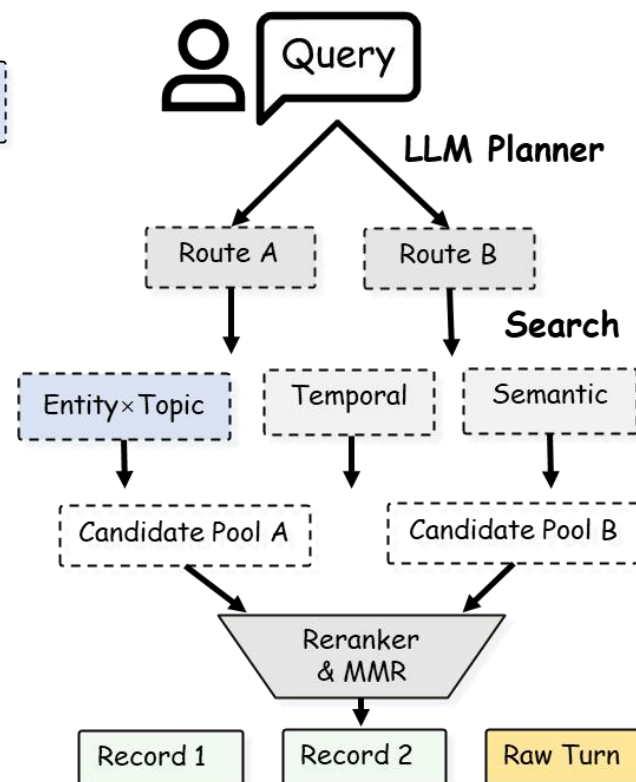
从自身运行轨迹中**自学习**更合适的记忆策略，使得长期记忆能随真实任务场景持续适配



记忆构建过程



记忆行为自进化



记忆检索过程

语义切分

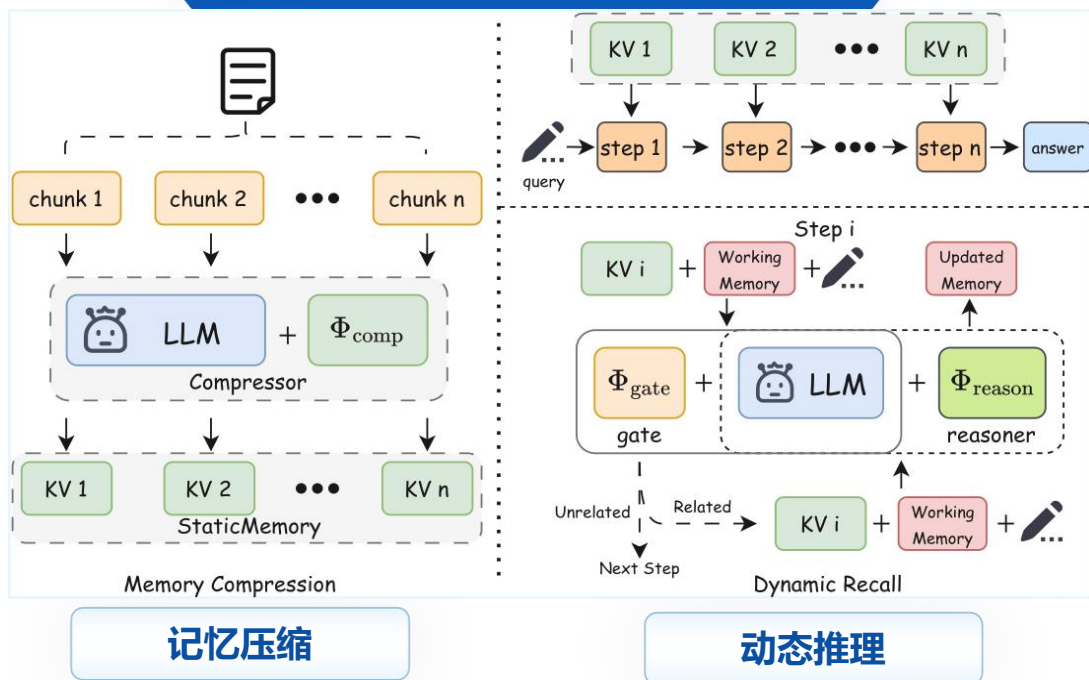
多元组织

多路检索

验证进化

构建了长短记忆协同的大模型LycheeMemory-7B。完成记忆构建、更新、推理等，将原始交互流转化为记忆单元，**实现了百万级长文推理的高准确率，已开源Huggingface**

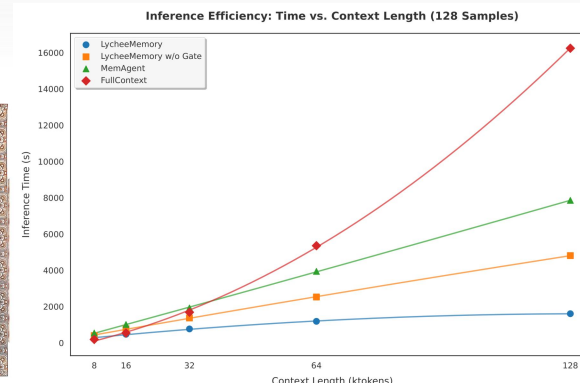
### 压缩记忆与动态推理



### 长文推理能力几乎无损外推至1.75M Token

| Model                                        | 7K                    | 14K                   | 28K                   | 56K                   | 112K                  | 224K                  | 448K                  | 896K                   | 1.75M                  |
|----------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|------------------------|
| QwenLong-L1-32B (Wan et al., 2025)           | 72.66                 | 75.00                 | 72.66                 | 60.94                 | 31.25                 | 17.19                 | 13.28                 | 11.72                  | OOM                    |
| Qwen2.5-Instruct-14B-1M (Yang et al., 2025a) | 60.16                 | 60.94                 | 50.00                 | 57.03                 | 50.00                 | 37.50                 | 8.59                  | 0.00                   | OOM                    |
| Qwen2.5-Instruct-7B-1M (Yang et al., 2025a)  | 61.72                 | 56.25                 | 53.91                 | 55.47                 | 51.56                 | 33.59                 | 12.50                 | 0.00                   | OOM                    |
| DS-Distill-Qwen-32B (Guo et al., 2025)       | 70.31                 | 66.41                 | 65.62                 | 46.88                 | 23.44                 | 13.28                 | 7.81                  | 7.03                   | OOM                    |
| DS-Distill-Qwen-14B (Guo et al., 2025)       | 64.06                 | 64.84                 | 57.03                 | 40.62                 | 14.84                 | 8.59                  | 3.12                  | 6.25                   | OOM                    |
| DS-Distill-Qwen-7B (Guo et al., 2025)        | 30.47                 | 12.50                 | 3.12                  | 0.00                  | 0.00                  | 0.78                  | 0.00                  | 0.00                   | OOM                    |
| RAG + Qwen2.5-7B-Instruct                    | 67.19                 | 66.41                 | 66.41                 | 67.19                 | 64.84                 | 64.06                 | 62.5                  | 61.72                  | 62.38                  |
| Search-R1 (Jin et al., 2025)                 | 72.66                 | 71.88                 | 67.71                 | 73.96                 | 66.67                 | 62.5                  | 64.58                 | 67.71                  | 67.19                  |
| RL-MemAgent-7B (Yu et al., 2025)             | <b>82.03</b>          | 79.69                 | 78.91                 | 77.34                 | 79.69                 | 72.66                 | 74.22                 | <b>76.56</b>           | 75.78                  |
| LycheeMemory-7B (ours)                       | 77.34 <sub>1.0x</sub> | 76.56 <sub>1.2x</sub> | 75.00 <sub>1.6x</sub> | 76.56 <sub>2.5x</sub> | 75.78 <sub>3.5x</sub> | 73.44 <sub>5.0x</sub> | 74.22 <sub>9.7x</sub> | 72.66 <sub>17.7x</sub> | 71.09 <sub>28.2x</sub> |
| LycheeMemory-7B w/o Gate (ours)              | 80.47                 | <b>81.25</b>          | <b>82.03</b>          | <b>81.25</b>          | <b>80.47</b>          | <b>79.69</b>          | <b>75.00</b>          | 75.78                  | <b>78.12</b>           |

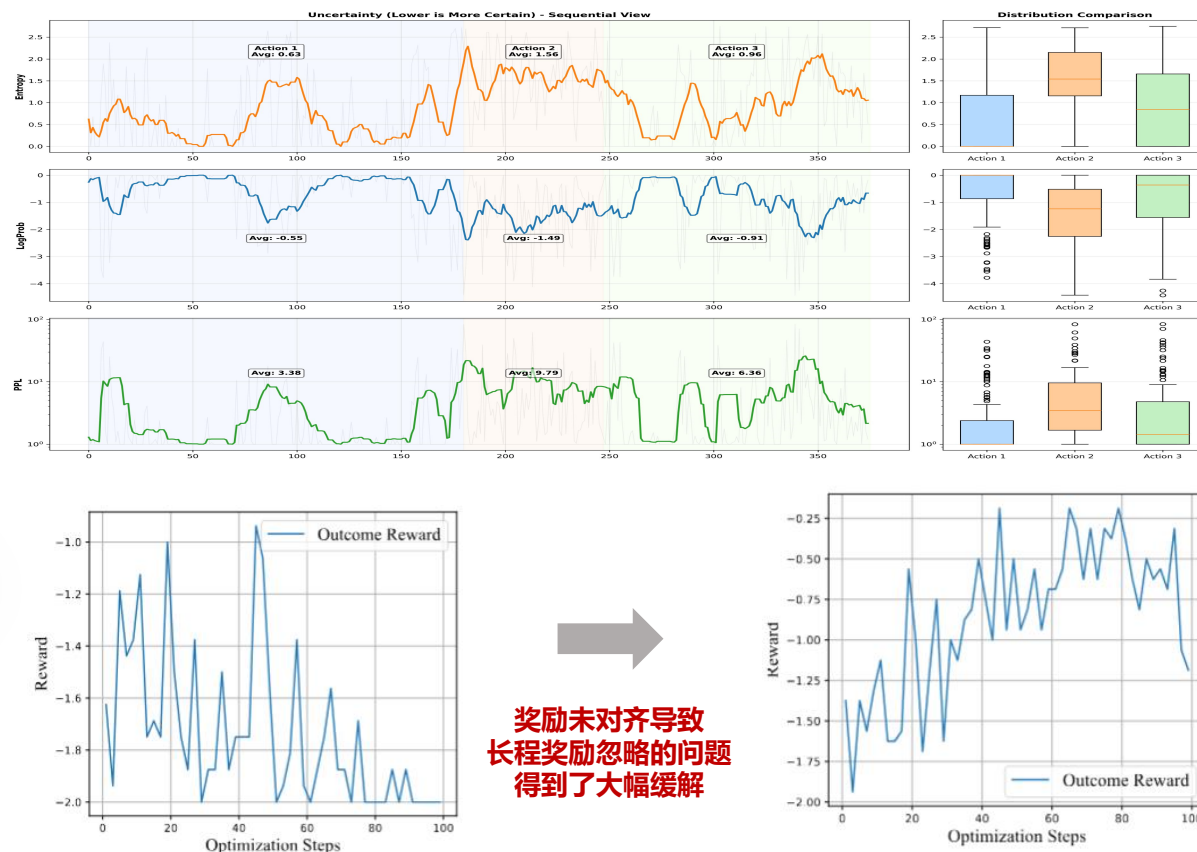
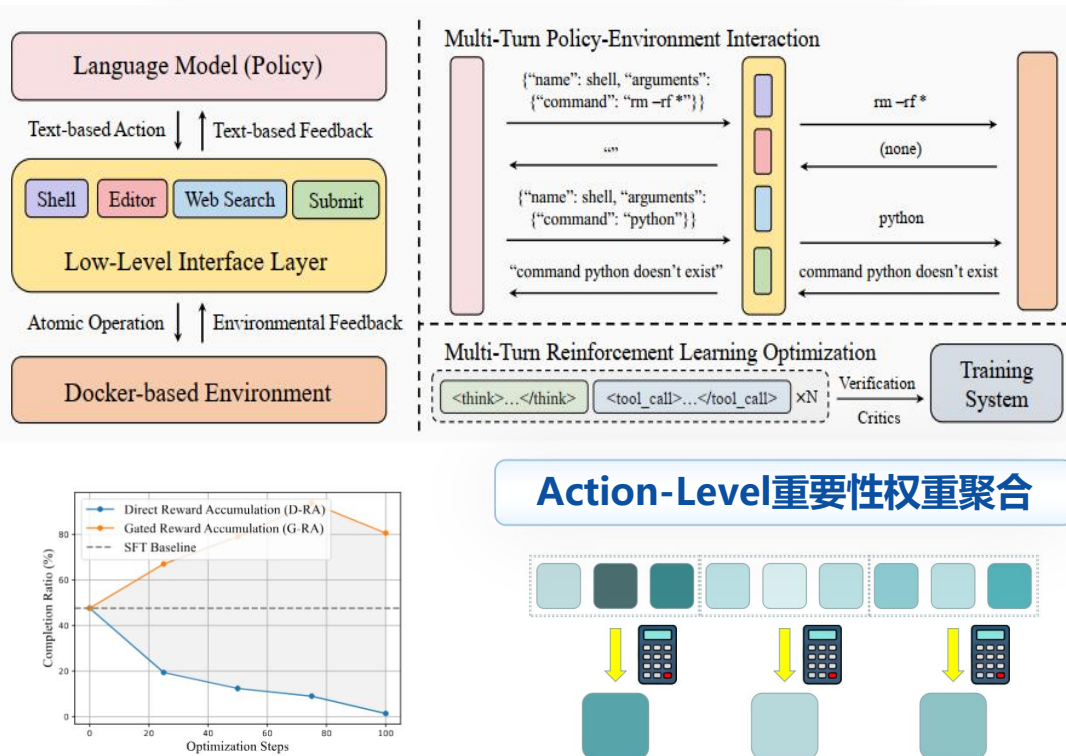
### 推理速度是MemAgent的6倍，全上下文的10倍





针对长程交互训练中**方差高、不稳定的问题**，提出了**门控奖励集成G-RA和多轮记忆生成策略优化AAPO方法**，**平衡了过程奖励和长程奖励**，最终提升了推理效果和训练稳定性

### 多粒度奖励驱动的深度强化学习



针对异步RL策略分布漂移与训练精度跷跷板等问题，提出了双轨去噪校准强化学习框架 DePO-REAL，显著提升了训练效率，稳定提高了模型精度，同时增强了多种任务能力

### 双轨去噪校准强化学习框架



| 方法        | AIME2025、LiveCodeBench、Arena Hard (步数/准确率) | 对比           |
|-----------|--------------------------------------------|--------------|
| 基线        | 300/27.7                                   | 训练步数减少       |
| SOTA      | 260/32.9                                   | 数学、代码、通用水平第一 |
| DePo-REAL | 100/34.1                                   |              |

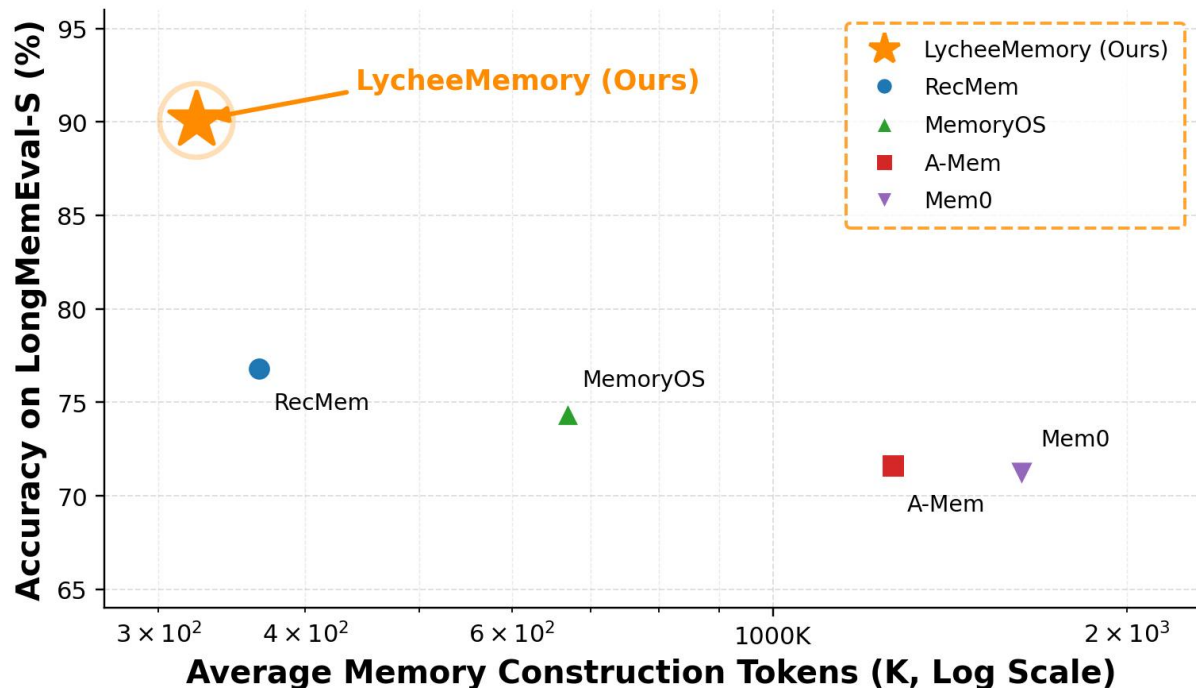
- 减少训练步数60%以上、大幅提升模型精度
- 超越现阶段多任务RL学习SOTA指标
- 模块化设计，代码少于200行，依赖少、可复现

LycheeMemory 在**显著降低记忆构建 Token 消耗**的同时  
在 LoCoMo 和 LongMemEval-S 上取得**更高的回答准确率**

LoCoMo

| 方法           | 准确率 (%)     | 构建<br>Token (K) | 查询<br>Token (K) |
|--------------|-------------|-----------------|-----------------|
| MemoryOS     | 67.6        | 400.7           | 5.04            |
| A-Mem        | <b>68.8</b> | <b>1459.9</b>   | <b>5.56</b>     |
| Mem0         | 62.9        | 1520.8          | 2.11            |
| LycheeMemory | <b>89.2</b> | <b>204.1</b>    | <b>4.01</b>     |

LongMemEval-S





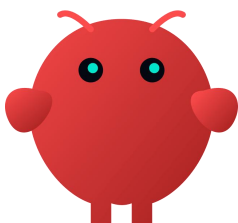
在 **PinchBench** 测评中，接入 LycheeMemory 的 OpenClaw 实现了更优越的性能  
重点不只是 “能记住”，而是在有限 Token 预算下 “**找得准、用得上**”

### 关键结果

- ❖ 整体评分提升 **6%**
- ❖ Token 消耗平均下降 **71.3%**
- ❖ 总使用成本降低 **54.8%**
- ❖ Hard 最高分数 **91.2%**



每投入 1 元记忆检索成本，最高可节省约 54.8 元大模型推理成本，ROI 达 5480%



## OpenClaw



## Claude



## HERMES AGENT

- **多平台插件接入**：支持 OpenClaw、Claude Code 和 Hermes 等 Agent Runtime，通过插件将长期记忆能力接入真实交互流程。
- **smart\_search 统一封装**：插件默认调用 smart\_search，将多层检索、重排、压缩与上下文构造封装为统一记忆召回接口。
- **Hook 生命周期管理**：通过会话开始、消息追加、会话结束等 hook，自动完成对话写入、记忆巩固与状态同步，减少手动调用成本。

通过华为OP专项的合作，我们记忆方法已成功应用到了Pangu-Embedded-7B

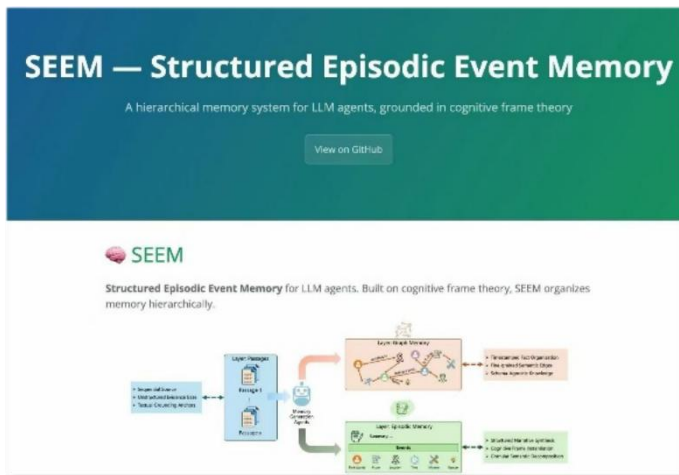
| Method                              | LoCoMo       |              |              | LongMemEval  |              |              |
|-------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                                     | F1           | EM           | $J$          | F1           | EM           | Acc.         |
| Context (Relevant Only)             | 38.60        | 11.40        | 69.28        | 22.60        | 3.20         | 68.60        |
| NV-Embed-v2 (Lee et al., 2025)      | 46.10        | 29.10        | 51.56        | 16.53        | 2.00         | 57.20        |
| <i>Memory-based Frameworks</i>      |              |              |              |              |              |              |
| HippoRAG 2 (Gutiérrez et al., 2025) | 44.60        | 27.90        | 50.96        | 34.82        | 25.10        | 46.59        |
| SEEM (Ours)                         | <b>48.60</b> | <b>30.80</b> | <b>56.11</b> | <b>45.50</b> | <b>29.60</b> | <b>60.80</b> |

Table 3: Performance comparison on LoCoMo and LongMemEval using the Pangu-Embedded-7B backbone. The best results are highlighted in **bold**.

构建了高信息密度与叙事连贯性的情节记忆，完全国产算力落地于openPangu模型  
实现了超越复杂基线方案的长程深度推理，期待在openPangu 2.0进一步应用



## 开源OpenClaw Skill无缝接入智能体 workflow，提供即插即用的结构化记忆技能



### 图记忆层样例

◆ ascend 960  
 ——[is planned]——→ for 2027  
 ——[has]——→ twice the computing power  
 ——[has]——→ twice the interconnect ports  
 ——[has]——→ twice the memory bandwidth

### 情节事件帧样例

— Event 2 🕒 2026-03  
 | 👤 Huawei  
 | 🌐 Huawei held the global debut of its AI computing clusters  
 | 📍 MWC Barcelona  
 | ? to build a resilient computing foundation and create a new option for global customers  
 | 🔧 by emphasizing open-source and open collaboration

### 记忆对象样例

Memory #1 [c7b0255a495d...]  
 |  
 | 📄 Alibaba is advancing its AI capabilities through significant investments in infrastructure and R&D, including procuring NVIDIA GPUs and Huawei Ascend chips to build large-scale AI training clusters and diversify its compute supply chain. The company holds a 35.8% share of China's AI cloud market in H1 2025, leads in key segments, and has strong financial performance driven by AI demand, with partnerships to drive enterprise intelligent transformation.  
 |  
 | 🕒 2025-01-01 → 2025-07-01 👤 Alibaba, Alibaba Cloud  
 | 📊 6 chunks · 10 events · 24 facts

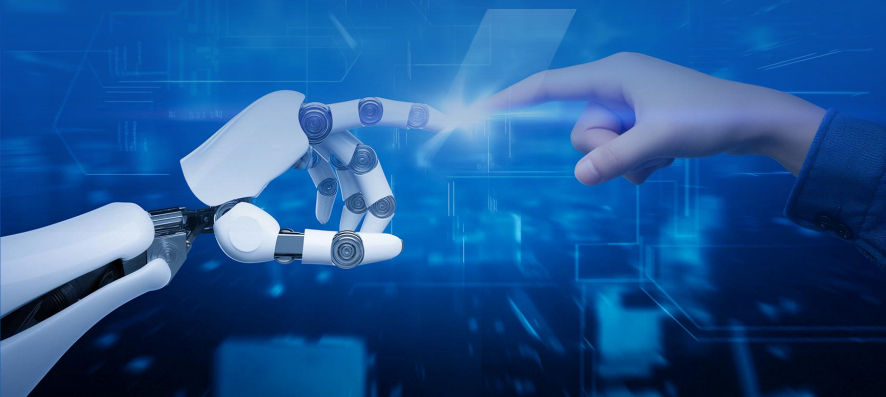
详见项目主页: <https://ryantoleco.github.io/seem-skill/>

# 主要内容

01 | 大模型记忆智能

02 | 智能体记忆工作

**03 | 相关工作与展望**



## DeepSeek-V4-Pro: 官方媒体发布与中央部委关注, 全网总阅读量超500万, 总转发量超5万

## 一、权威媒体发布

## 1 河套学院公众号首发

全球首发 | SLAI基于国产算力集群  
完成DeepSeek-V4-Pro全参数后训练

原创 深圳河套学院 深圳河套学院

2026年5月28日 19:56 广东 32人



首发标题聚焦: 国产算力集群完成 DeepSeek-V4-Pro 全参数后训练

## 2 新华社视频报道



报道聚焦: 1.6万亿参数大模型后训练, 国产算力跑通关键一环



河套学院首发



新华社视频报道



深圳发布跟进

新华社  
XINHUA NEWS

1.6万亿参数 MoE

DeepSeek V4 Pro 后训练 /  
SFT 阶段性验证

昇腾910C集群

承接超大参数模型训练链路  
关键环节

权威传播链路

首发 · 新华社视频 ·  
深圳发布跟进

新华社、人民网、大公报、文汇报、深圳特区报、深视新闻、南华早报、深圳发布、证券时报、凤凰网等

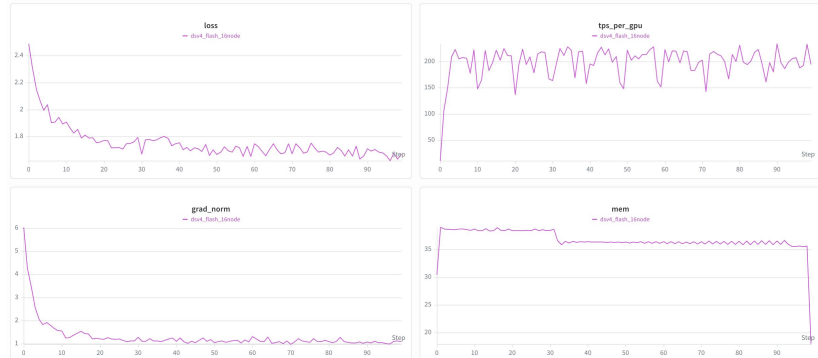


# 全球首发：基于国产算力集群完成的DeepSeek-V4-Pro全参数后训练工程实践，实现了V4 Flash/Pro的SFT/续训练。未来将依托国产算力底座，打造昇腾亲和的新一代记忆原生的模型架构

## 基于千卡国产算力完成的DeepSeek-V4-Pro全参数SFT

- V4-Flash mindspeed 续训 ✓
- V4-Flash torchtitan 续训 ✓
- V4-Flash 8K SFT 训练 ✓
- V4-Pro torchtitan 续训 ✓
- V4-Pro 8K SFT 训练 ✓

```
[rank0]:[titan] 2026-05-19 20:10:39,182 - root - INFO - [31mstep: 8 * [32mloss: 6.11353 * [38;2;180;60;0mgrad_norm: 14.7876 * [38;2;54;234;195mmemory: 48.82GiB(79.67%) * [34mtmps: 69 * [36mtflops: 61.51 * [35mmfu: 17.38% * [33mlapsed_time_per_step: 59.133s * [39m
[rank0]:[titan] 2026-05-19 20:10:39,185 - torchtitan_npu.models.deepseek_v4.infra.parallelize - INFO - indexer loss: 0.09184350818395615
[rank0]:[titan] 2026-05-19 20:11:37,677 - root - INFO - [31mstep: 9 * [32mloss: 6.07100 * [38;2;180;60;0mgrad_norm: 14.6897 * [38;2;54;234;195mmemory: 48.83GiB(79.70%) * [34mtmps: 70 * [36mtflops: 62.19 * [35mmfu: 17.57% * [33mlapsed_time_per_step: 58.491s * [39m
[rank0]:[titan] 2026-05-19 20:11:37,682 - torchtitan_npu.models.deepseek_v4.infra.parallelize - INFO - indexer loss: 0.09131671488285065
[rank0]:[titan] 2026-05-19 20:12:34,853 - root - INFO - [31mstep: 10 * [32mloss: 6.08417 * [38;2;180;60;0mgrad_norm: 14.4792 * [38;2;54;234;195mmemory: 48.82GiB(79.67%) * [34mtmps: 72 * [36mtflops: 63.62 * [35mmfu: 17.98% * [33mlapsed_time_per_step: 57.174s * [39m
[rank0]:[titan] 2026-05-19 20:12:34,857 - torchtitan_npu.models.deepseek_v4.infra.parallelize - INFO - indexer loss: 0.09163681417703629
[rank0]:[titan] 2026-05-19 20:13:31,542 - root - INFO - [31mstep: 11 * [32mloss: 6.05960 * [38;2;180;60;0mgrad_norm: 14.7472 * [38;2;54;234;195mmemory: 48.82GiB(79.67%) * [34mtmps: 72 * [36mtflops: 64.17 * [35mmfu: 18.13% * [33mlapsed_time_per_step: 56.684s * [39m
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```



训练Loss正常下降  
训练指标定时统计

## 算子优化&数据&评测：测评集初步筛选，持续构造中

| 测评集           | 样本数 | 测评工程/脚本 |
|---------------|-----|---------|
| ORGEval       | 394 | 构建完成    |
| NL4OPT        | 245 | 构建完成    |
| MAMO_EasyLP   | 642 | 构建完成    |
| MAMO_ComplexL | 203 | 构建完成    |
| IndustryOR    | 100 | 构建完成    |
| OptMath_166   | 166 | 构建完成    |
| OptiBench     | 605 | 构建完成    |

测评集/工程初版本  
构建完成

| Domain                  | Token Count | Percentage | Description                                       |
|-------------------------|-------------|------------|---------------------------------------------------|
| Nemotron CC High        | 1,468.3B    | 59.7%      | High quality CommonCrawl data                     |
| DCLM                    | 314.2B      | 12.8%      | DCLM baseline web content                         |
| RefineCode              | 279.4B      | 11.4%      | GitHub repositories (Academic Use Only)           |
| Nemotron CC Medium-High | 254.5B      | 10.3%      | Medium-high quality CommonCrawl data              |
| FineWeb Edu             | 117.4B      | 4.8%       | Educational web content                           |
| Chinese                 | 112.18B     | 4.6%       | Chinese general content                           |
| Reasoning QA            | 86.2B       | 3.5%       | Instruction-following and complex reasoning tasks |
| Math Web                | 68.3B       | 2.8%       | Mathematics and scientific content                |
| MegaMath                | 28.5B       | 1.2%       | Specialized mathematical collections              |
| Translation             | 1.61B       | 0.1%       | English-Chinese translation pairs                 |
| Total                   | 2,460.71B   | 100%       | Complete dataset                                  |

数据集续训数据  
2.6T->50B、专家  
构建数据，进行中

# 全球首发：基于国产算力集群完成的DeepSeek-V4-Pro全参数后训练工程实践，实现了V4 Flash/Pro的SFT/续训练。未来将依托国产算力底座，打造昇腾亲和的新一代记忆原生的模型架构

 深圳河套学院  
Shenzhen Loop Area Institute
SLAI<sup>2</sup> Training Platform Technical Report (Part I)

## SLAI T-Rex: Full-Parameter Post-training of the DeepSeek-V4 Family on Ascend SuperPOD

AI Training Platform Team, Shenzhen Loop Area Institute

Models Codes

### Abstract

Full-parameter post-training of trillion-parameter-scale Mixture-of-Experts (MoE) models introduces substantial system-level challenges for large-scale distributed training, including severe memory pressure, non-overlapped communication overhead, and inefficient kernel execution. While most large-scale LLM training systems are built around GPU-based clusters, this report presents an end-to-end optimization practice on the Ascend NPU SuperPOD. Using the DeepSeek-V4 model family as the target workload, we develop a hierarchical optimization framework spanning model-level parallelism, computation-communication orchestration, and low-level kernel execution. The resulting system achieves 34.22% Model FLOPs Utilization (MFU) with a 2.93x improvement over the open-source baseline recipe while maintaining training stability. Building on this optimized infrastructure, we further establish a Continued Pre-Training (CPT) and Supervised Fine-Tuning (SFT) workflow for complex Operations Research (OR) tasks. We refer to the integrated framework as SLAI T-Rex, which couples full-stack Ascend SuperPOD optimization with solver-grounded CPT-SFT specialization. OR modeling is selected as the target domain because it jointly requires mathematical formulation, structured reasoning, and code generation. Using DeepSeek-V4-Flash as the specialization platform, we develop OR-oriented CPT and SFT data pipelines that combine collected domain resources with solver-verified synthetic optimization documents. The resulting dataset contains 10K high-quality SFT samples spanning four task categories and three problem representations. The specialized model achieves the highest average zero-shot Pass@1 score among the evaluated models, reaching 71.81% and outperforming GPT-5.4-Mini and the base DeepSeek-V4-Flash model by 3.98 and 11.27 percentage points, respectively. Further evaluation indicates that CPT establishes transferable OR-domain modeling priors, improving both solver-facing feasibility and structural equivalence after SFT. Overall, this work demonstrates a full-stack pathway from efficient trillion-parameter model post-training on Ascend infrastructure to domain-specialized Flash models for solver-grounded mathematical modeling, advancing frontier-model systems for complex reasoning.

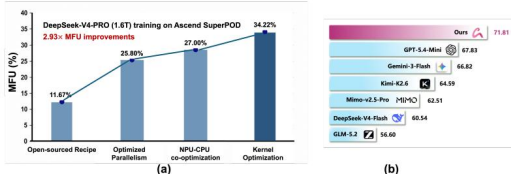


Figure 1: (a) MFU improvements during DeepSeek-V4-Pro (1.6T) training on Ascend SuperPOD. Starting from open-sourced recipes, optimized parallelism, NPU-CPU co-optimization, and kernel optimization increase MFU from 11.67% to 34.22%. (b) Overall accuracy, computed as the mean of NL4OPT (Ramamonjison et al., 2022), OptiBench (Yang et al., 2025c), Bench4Opt (B4O)-Feasible (Wang et al., 2025b), and Bench4Opt (B4O)-ORGEval (Wang et al., 2025b). SLAI T-Rex achieves the highest overall score while using a smaller disclosed activated-parameter budget.

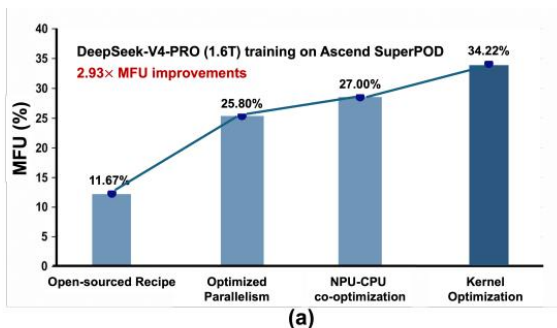
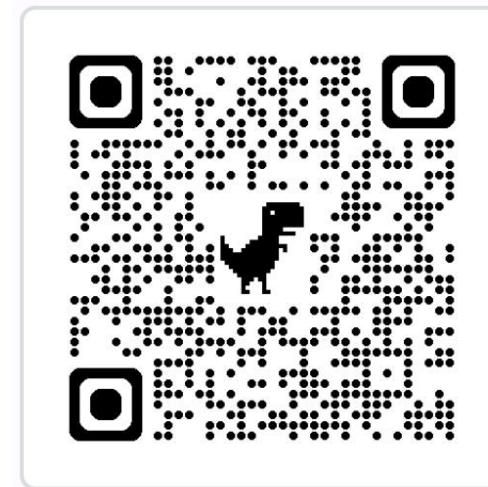
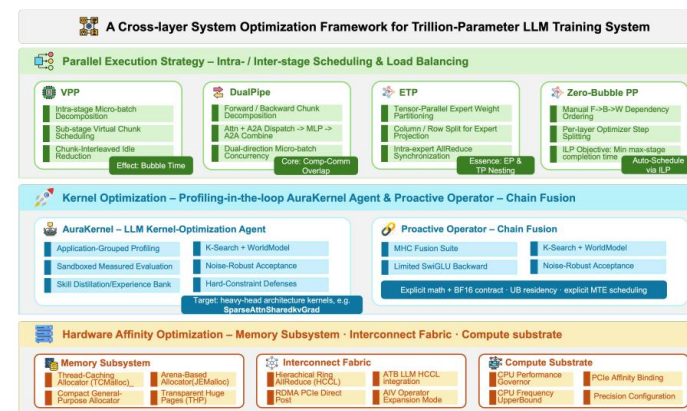


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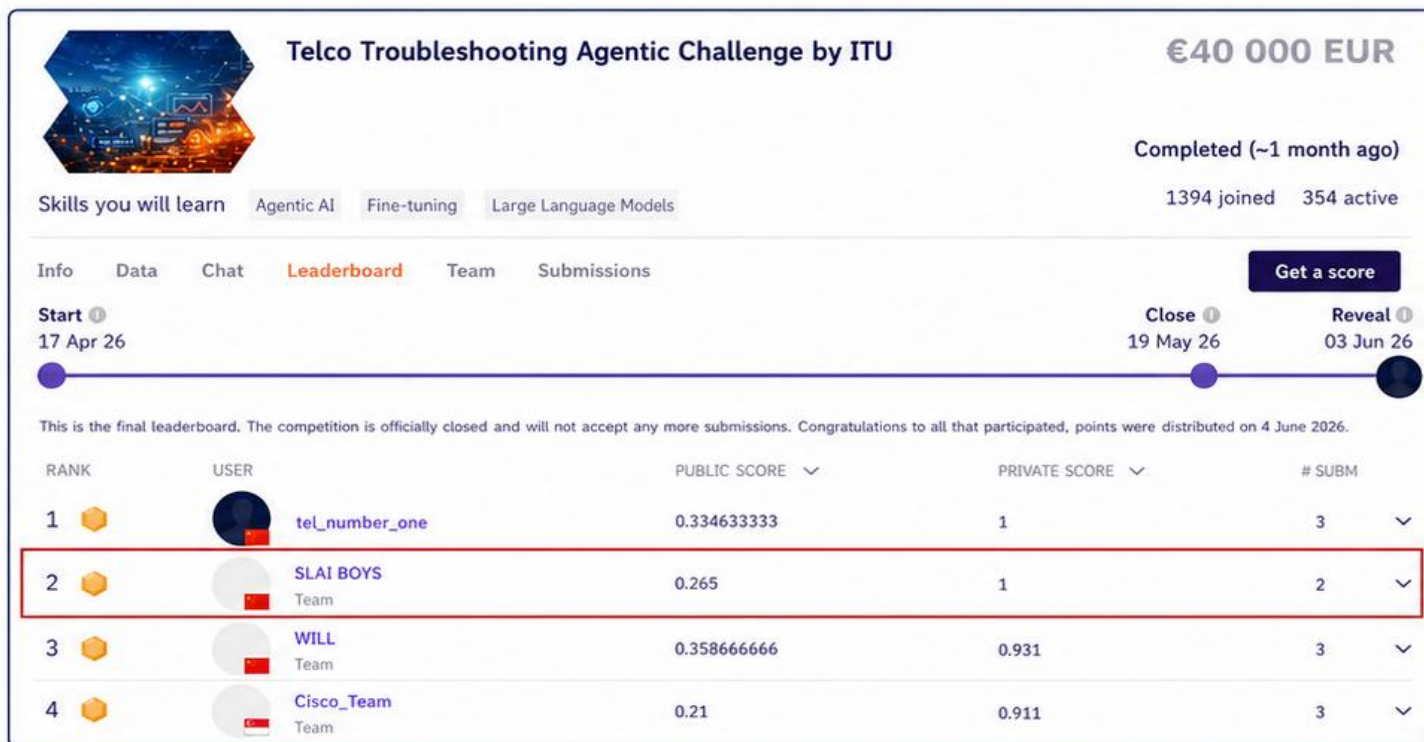
## MFU 提升趋势 (%)

整体提升约 2.91×





## AI训练平台学生团队在国际电信联盟ITU举办的全球电信AI AGENT挑战赛中获世界冠军，并受邀报告



官方最终榜单：SLAI BOYS 位列 Track B Phase 3 Final 第一；Cisco\_Team 等专业队伍同场竞争。



### 1. 国际公信力

ITU AI for Good 相关平台推动，联合 GSMA、ETSI、IEEE GenAINet、TM Forum 等国际组织。

### 2. 赛道约束

开放式 IP 网络排障，多厂商 CLI、隐藏网络环境、多步骤诊断与严格提交格式。

### 3. 强对手验证

超过 Cisco\_Team 等专业队伍，说明结果经过国际公开赛道竞争检验。

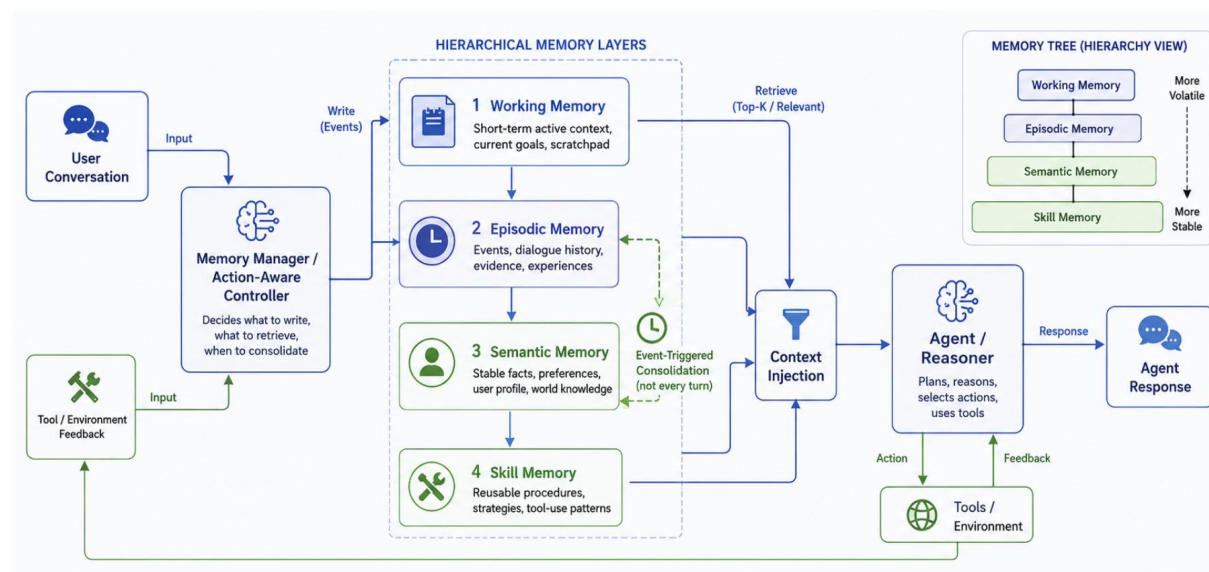
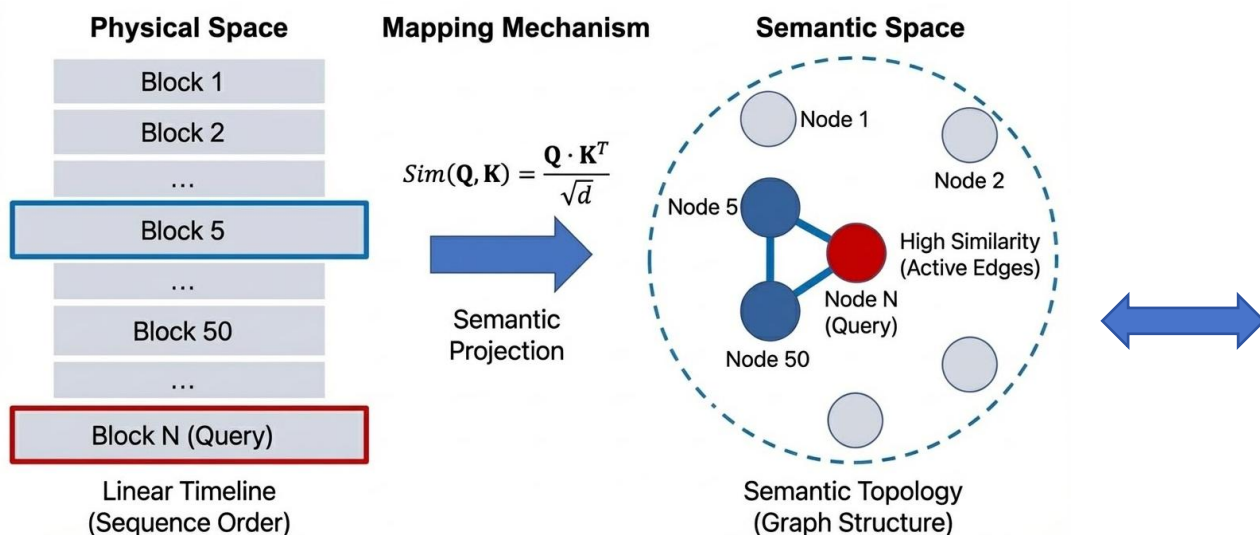
### 4. 平台作用

AI 训练平台把学生训练、模型训练、工程调试、真实评测串成闭环。

### 结论：

平台训练出的模型和队伍，能够完成真实复杂系统问题解决，并在全球公开赛道取得世界冠军。

# 进一步提高对记忆机理的认知，不仅是工程实现，挖掘关键科学问题 从面向 Agent 的长期记忆系统设计，到记忆原生大模型新型架构的构建



面向无限长序列的记忆原生的动态图索引新型架构

降低长文本推理开销，提升大模型处理效率

行动感知驱动的智能体层次化记忆

让记忆真正参与写入、检索、巩固和注入全周期



# Memory Scaling: 大模型/智能体持续学习机制





**谢谢！**

**敬请各位专家批评指正！**